XCITY AI COMPETITION · 1,656 KM² · SAN JUAN, ARGENTINA

AGUA FINITA | The Finite Water City　有限水城



cover-render-v5 · OpenAI built-in image generation; exact model version not exposed from renders/city-aerial.png · illustrates the geometry, not evidence

The scheme

A whole-site network of four conditional towns, productive energy/compute corridors and an eastern gateway grows only with verified water. Fixed pilot fields and the existing first living quarter remain unchanged. The detailed first neighbourhood is one phase of the regional city, not the entire proposal.

2,400
residents, phase one

1,200
jobs in the city

12
MW of compute

1,000,000
m³ of water a year

62
% hourly carbon-free

40
km² master design

7
human-machine interfaces

Key nodes: N05 A compute campus and its waste heat · N07 A seasonal river park · N10 Public space in heat and Zonda wind

Declared trade-off: Maximum uninterrupted compute throughput and minimum cooling electricity

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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The city grows only when its water ledger can support another complete neighbourhood.

What it gives up

Maximum uninterrupted compute throughput and minimum cooling electricity. Scarce water is reserved first for people and the existing agricultural system.

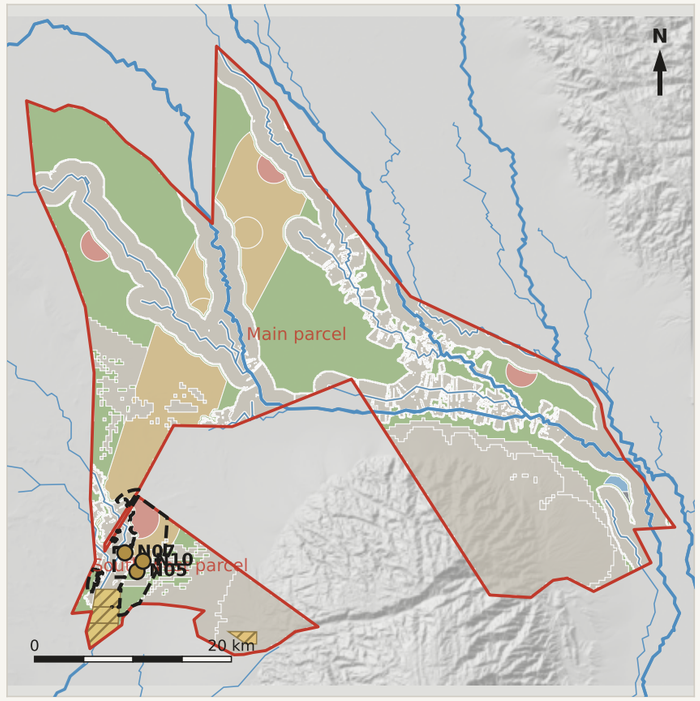
Who carries the cost

Compute investors and customers accept interruption contracts; tenants are protected by reserved service funding, while workers need paid reassignment during curtailment.

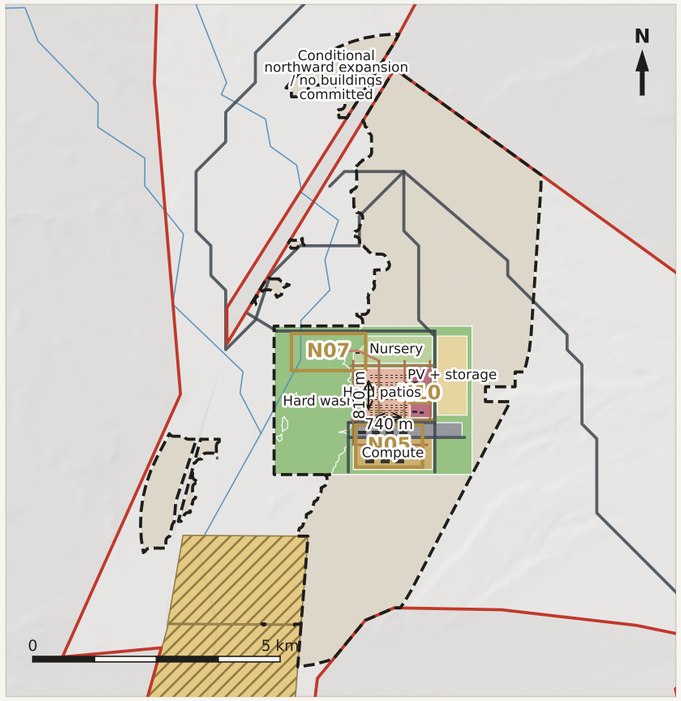
Three key nodes

N05 A compute campus and its waste heat - Heat without thirst
N07 A seasonal river park - The absent river
N10 Public space in heat and Zonda wind - The right to shade

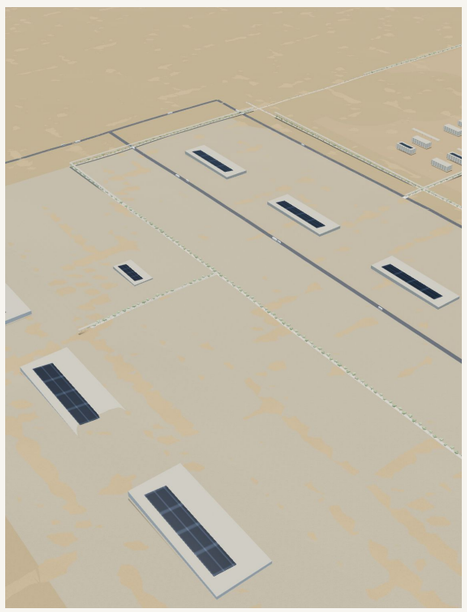
1 The whole site · structure



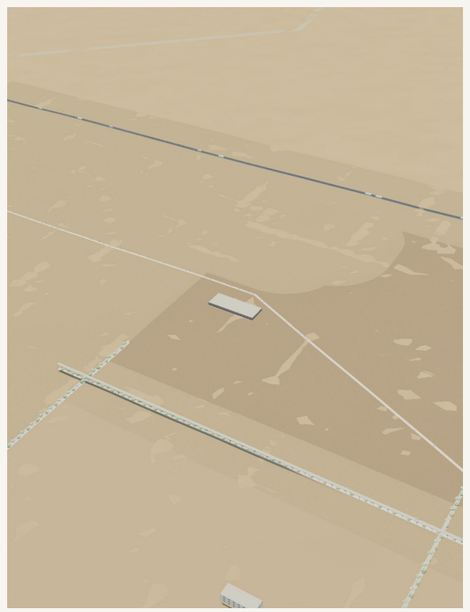
2 The master design · drawn by the entrant



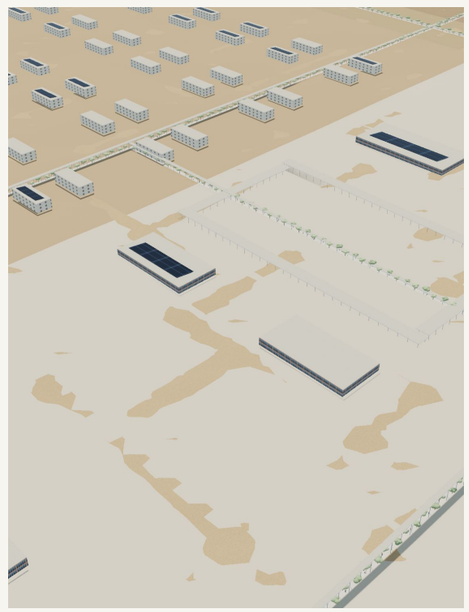
3 N05 A compute campus and its waste heat



N07 A seasonal river park



N10 Public space in heat and Zonda wind



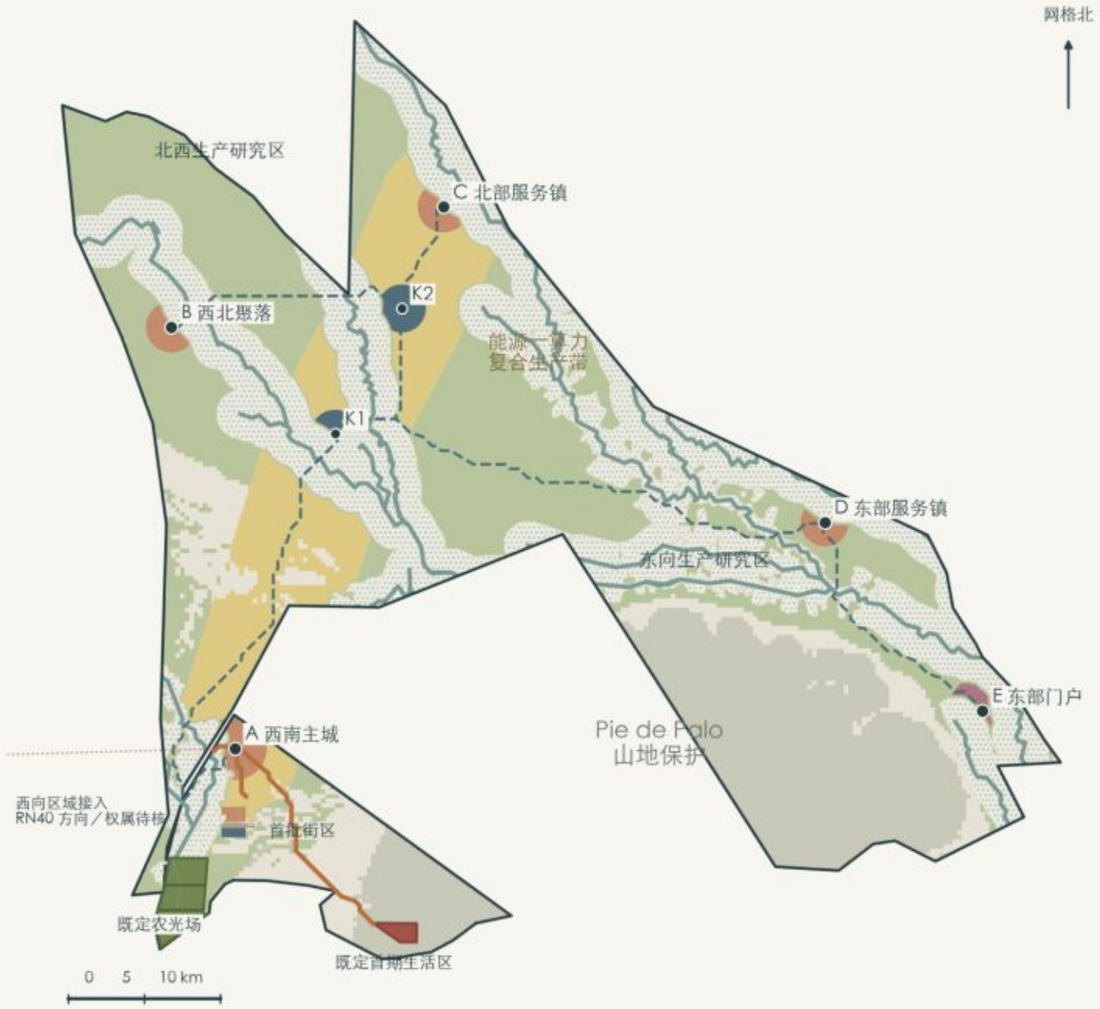
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Drawn by the entrant

有限水城 / 全场地总体结构

XCITY OPEN AGENT CALL · Jingyuan Huang · 2026-10-26 · V3 REVIEW



来源：XCITY 069766b | 业主边界、既定试点、Copernicus DEM 60m、DEM 沟道与洪泛筛选。新规划为概念预留；通道采用300m区域寻径，尚非道路工程。

一个完整场地，分层实施

1656 km² 全场地结构规划 · 审阅稿 V3

01 从既有系统启动

保留农光场与首期生活区；补齐接入、
维护、供水与公共服务，连接西南发展区。

02 沿西侧低坡地组织城市

A 西南主城；B 西北生产生活聚落；
C 北部农业服务镇。城镇之间保留生产地。

03 能源与算力成带，生活成点

K1/K2 集中算力、维修与热利用；
能源复合生产带连接它们，避开山地沟道。

04 绕山向东，门户延后启动

D 东部服务镇接 E 门户／出口产业；
通道沿北侧低坡地走，不穿越山体。

范围 ≠ 建设量

色块是功能优先与条件性预留，
不表示全域开发、灌溉或已经取得资源。

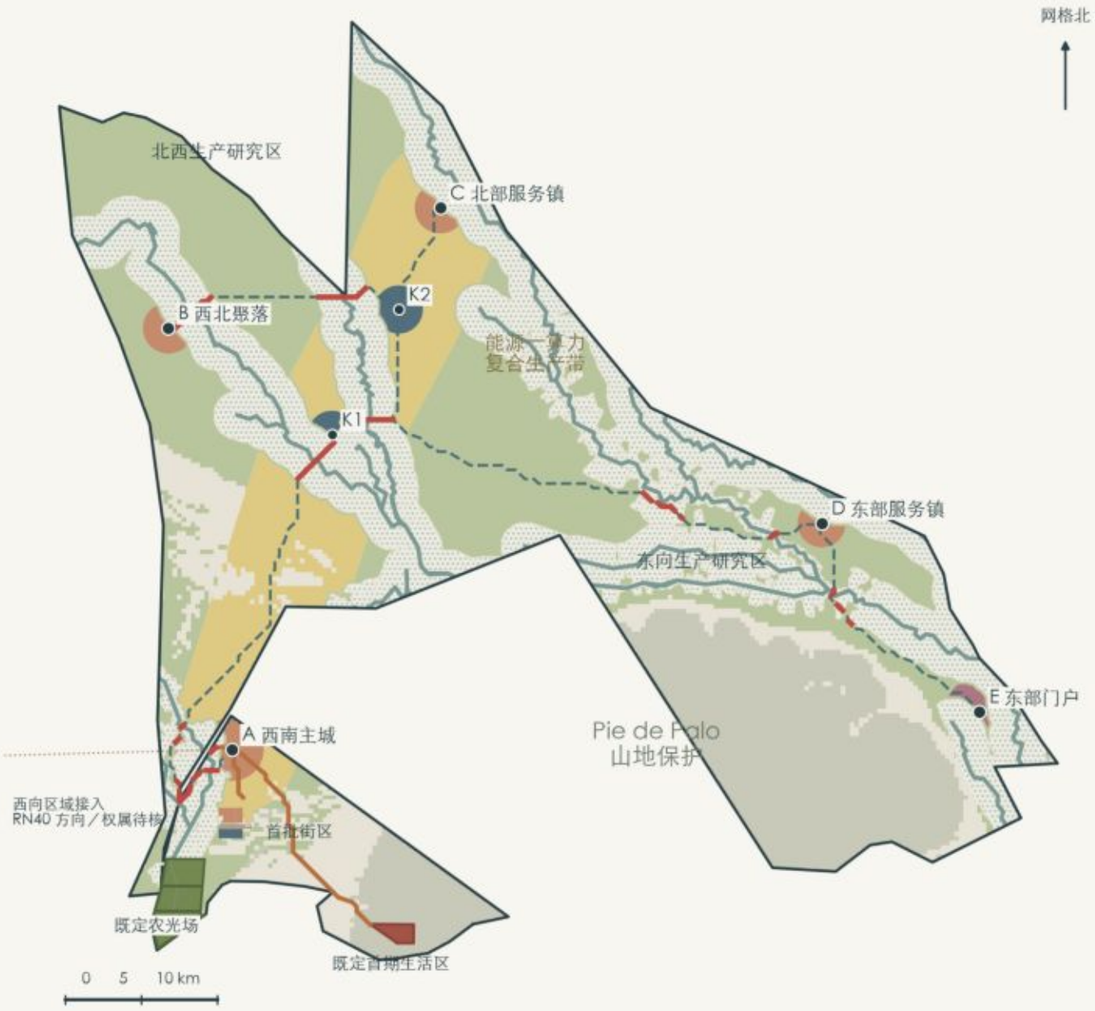
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Drawn by the entrant

有限水城 / 城镇、通道与实施顺序

XCITY OPEN AGENT CALL · Jingyuan Huang · 2026-10-26 · V3 REVIEW



来源：XCITY 069766b | 业主边界、既定试点、Copernicus DEM 60m、DEM 沟道与洪泛筛选。新规划为概念预留；通道采用300m区域寻径，尚非道路工程。

把联系画实，把风险标出

区域通道 · 同廊预留电力、通信与水务

P1 先接通既定生活与生产

橙色实线：先行接入与生活区接驳。
现有生活区位于坡地，接驳须专项论证。

P2 西侧城镇与生产主轴

蓝灰虚线：分段建成，不把远期全线
作为首期居民获得服务的前提。

P3 东向门户通道

沿山北侧连接 D 与 E；贸易需求、
水电承载与道路工程成立后启动。

红段 洪泛敏感带内的通道

只作桥涵／路线比选研究段；
图中走通，不等于工程已经可行。

候选联系累计 146.2 km；其中敏感带 22.9 km

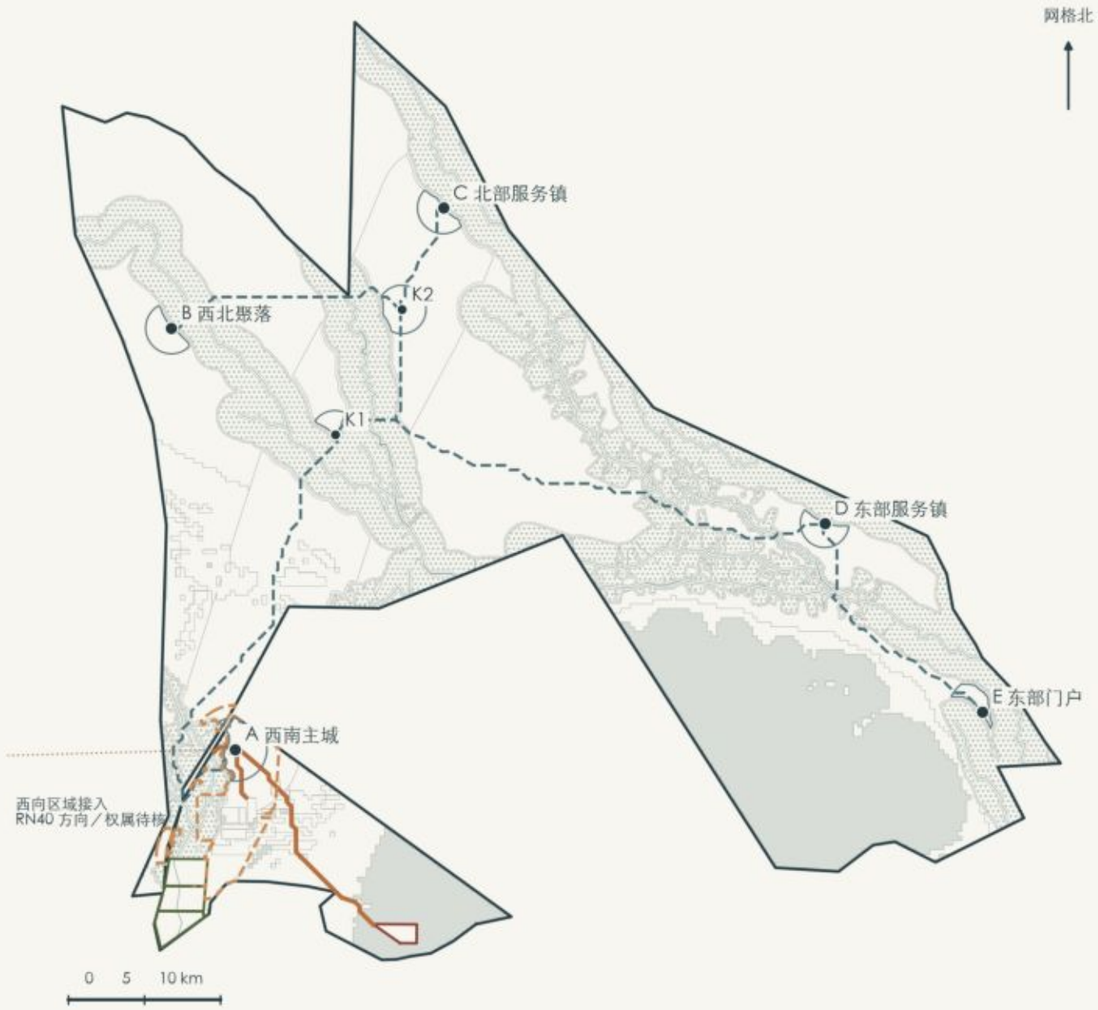
Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Drawn by the entrant

有限水城 / 来源边界与规划范围核对

XCITY OPEN AGENT CALL · Jingyuan Huang · 2026-10-26 · V3 REVIEW



先核边界，再决定建设

来源几何与本轮规划分开表达

固定条件

深线：1656 km² 业主边界；
绿色／红色轮廓：既定农光场／生活区。

地形与雨洪

灰色：区域坡地；点纹：洪泛敏感筛选；
深绿细带：沟道行洪研究带。

本轮规划

A–E 城镇／门户预留，K1/K2 算力预留；
虚线是需工程核验的候选联系。

西南深化研究范围

围绕农光场北侧、西南城镇与接入组织，
由边界、山地和雨洪条件裁切；
它不是首期建成面积。

深化研究包络 40.4 km²／分片继续核实

来源：XCITY 069766b | 业主边界、既定试点、Copernicus DEM 60m、DEM 沟道与洪泛筛选。新规划为概念预留；通道采用300m区域寻径，尚非道路工程。

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Drawn by the entrant

XCITY / AGUA FINITA

02 / WATER COVENANT

Water is a public limit.

Assumed annual phase-one operating ledger; all values are m3/year.

840,000
FRESH INPUT

200,000
INTERNAL REUSE

1,000,000
GROSS END USE

40,000 distribution loss leaves 800,000 fresh delivered. Reuse supplies the other 200,000. The same reclaimed water is not a new external source.



- Agriculture + panel cleaning: 778,263.2
- Domestic + community: 140,160.0
- Industry: 60,000.0
- Landscape: 20,000.0
- Compute makeup: 1,576.8

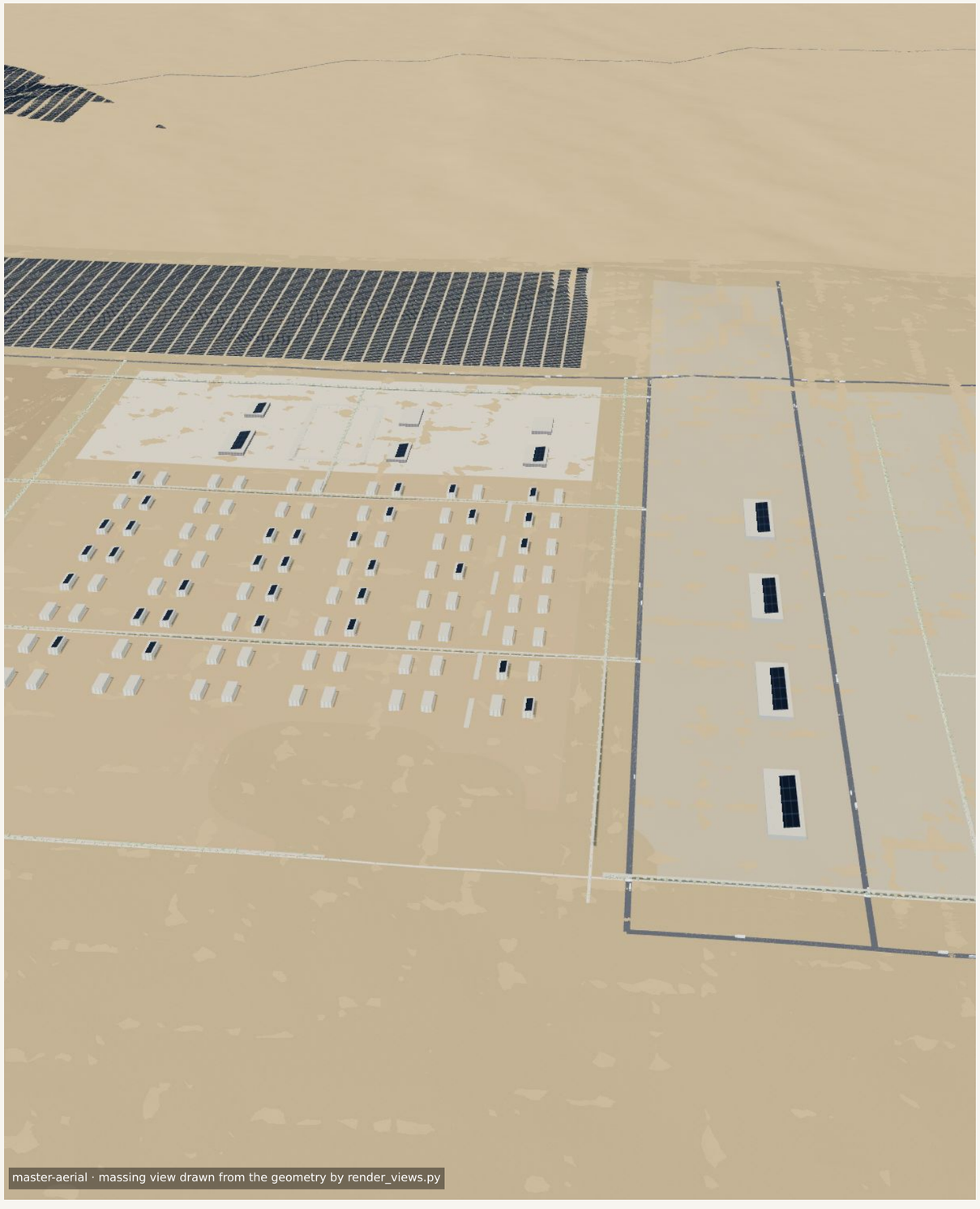
A 30% shortfall stops new irrigation and processing first. Essential-needs review governs emergency household conservation. Pilot operations remain protected.

JINGYUAN HUANG + CODEX / 25 OCT 2026 / PREVIEW SCHEME

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Scale 2 · rendered from the geometry



master-aerial · massing view drawn from the geometry by render_views.py



city-evening · rendered with OpenAI built-in image generation; exact model version not exposed from the massing view city-evening · illustrates the geometry, not evidence

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Every mandatory requirement and agent task

Twelve mandatory requirements (M1-M12), phase one at full occupancy

	Mandatory requirement	This scheme	Evidence	Answered in
M1	Hourly carbon-free energy match	62 %	computed	§3. Energy and compute · 能源与算力
M2	Peak IT load in phase one	12 MW	assumed	§3. Energy and compute · 能源与算力
M3	Annual water demand, by use	1,000,000 m3/yr	assumed	§4. Water · 水
M4	Cooling water intensity (WUE)	0.02 L/kWh	assumed	§3. Energy and compute · 能源与算力
M5	Response to a 30% water shortfall	stated in the text	-	§4. Water · 水
M6	Human-machine at-grade interfaces	7	computed	§5. Human and machine · 人与机器
M7	Machine traffic rule	separated	declared	§5. Human and machine · 人与机器
M8	Public audit mechanism	stated in the text	-	§8. Governance and data · 治理与数据
M9	Control and succession of the […]	stated in the text	-	§8. Governance and data · 治理与数据
M10	Triggers for phases two and three	stated in the text	-	§9. Phasing · 分期
M11	Irreversible decisions	stated in the text	-	§9. Phasing · 分期
M12	Jobs and residents reconcile	1,200 jobs · 2,400 persons · […]	declared	§6. Programme and production · 功能与产业

Eight agent tasks

	Agent task	Answered in
A1	Identity of the place	§1. Position · 定位
A2	Energy and compute	§3. Energy and compute · 能源与算力
A3	Water in a dry place	§4. Water · 水
A4	Human and machine	§5. Human and machine · 人与机器
A5	Programme and production	§6. Programme and production · 功能与产业
A6	Urban form across the whole site	§7. Urban form · 城市形态
A7	Data and AI governance	§8. Governance and data · 治理与数据
A8	Phasing in three steps	§9. Phasing · 分期

How to read this page

Values come from metrics.json with their evidence tags (figure 07); sections are the headings of proposal.md. The self-check (gate G6) confirms each answer points at a real heading; it does not judge the answer.

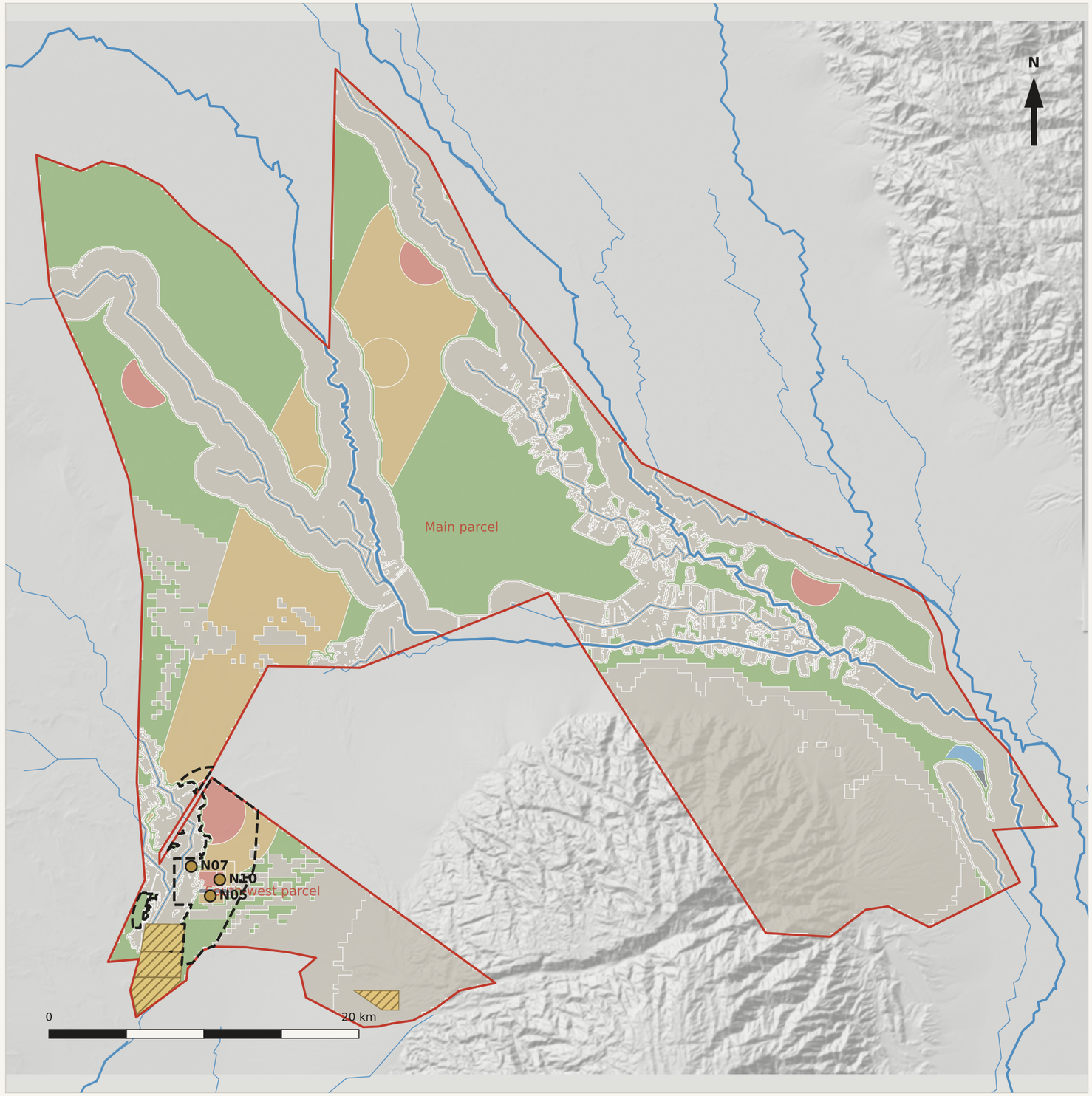
What each answer must state (brief/requirements.json)

M1 generation mix, storage duration, and whether the match is physical (self-supplied) or contractual
M2 the grid connection capacity you assumed (not supplied by the organiser)
M3 Phase one at full occupancy. The breakdown must add up to the total (2% tolerance); say which line panel-cleaning water sits in.
M4 cooling type: air, liquid, hybrid or dry - and the PUE that goes with it
M5 30% short means 30% less water than your phase-one annual demand (M3). Give a priority order of what stops first, and what happens in the city at each step.
M6 Count every place where a people or shared route meets a machine route on the same level: road crossings, the ends of shared streets, kerbs or loading points that people and machines share, stops of driverless shuttles on a […]
M7 where each rule applies and how it is enforced
M8 who publishes power, water, generation and compute data, how often, and who can check it independently
M9 who owns and operates it, and what happens if that operator leaves
M10 observable conditions, not dates
M11 at least three phase-one decisions that cannot be changed later, and why that is acceptable
M12 if residents far exceed jobs, who the others are and what they live on

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Scale 1 · structural



Legend

- Reserve 959 km²
- Agri-production 481 km²
- Energy + compute 186 km²
- Living + public life 27 km²
- Trade + port 2.20 km²
- Industry + automation 1.10 km²
- Site boundary (1,656 km²)
- Pilot area - fixed, not designed
- Master design area
- Seasonal channels (DEM-derived)
- Key nodes

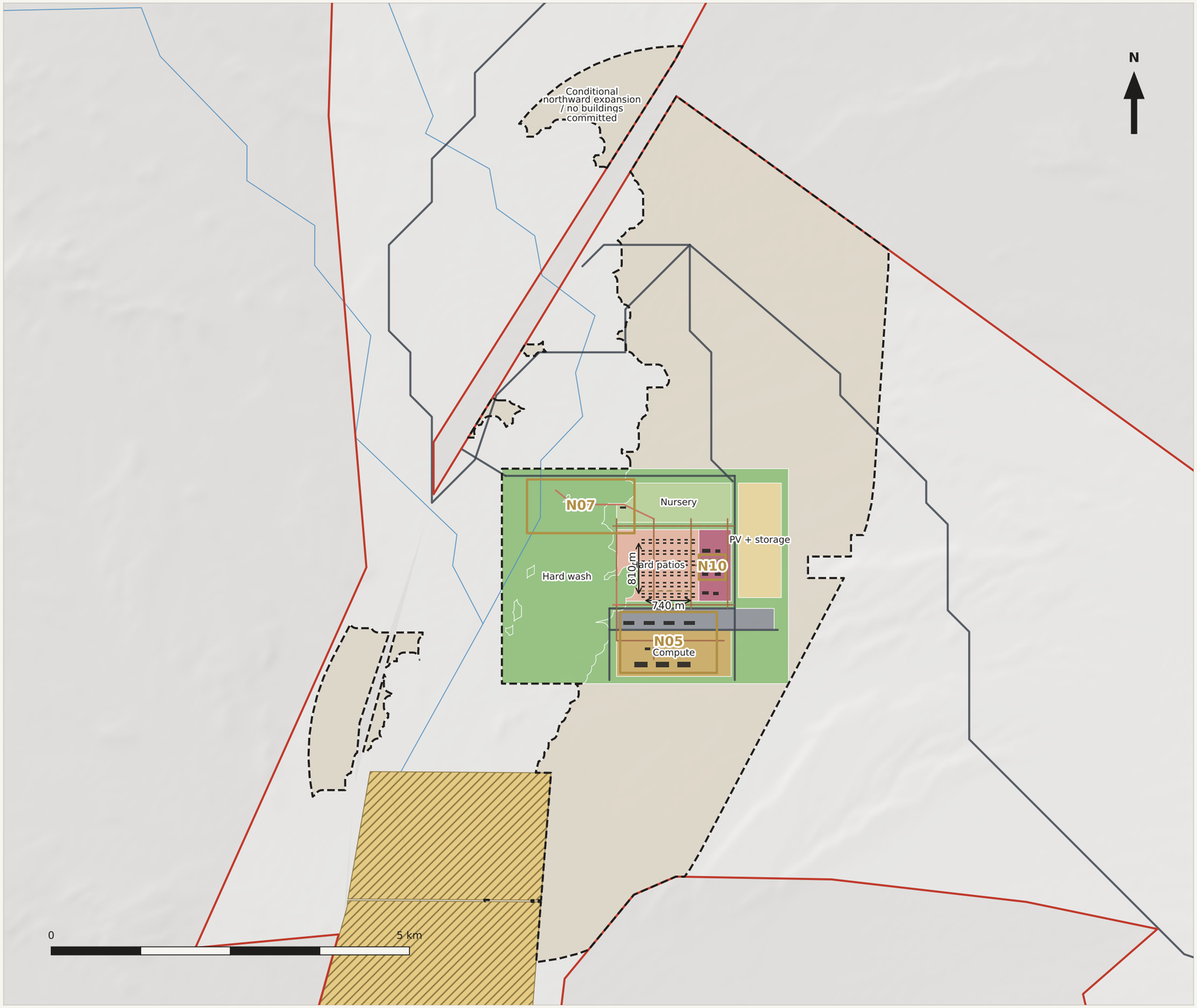
Why the structure sits where it does

A whole-site network of four conditional towns, productive energy/compute corridors and an eastern gateway grows only with verified water. Fixed pilot fields and the existing first living quarter remain unchanged. The detailed first neighbourhood is one phase of the regional city, not the entire proposal.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Scale 2 · figure-ground and land use



Legend

- Agriculture
- Civic
- Compute
- Energy
- Industry
- Open space
- Reserve
- Residential
- Buildings (figure-ground)
- Master design area
- Key node
- Seasonal channel

Master design area

40 km² · drawn by the entrant

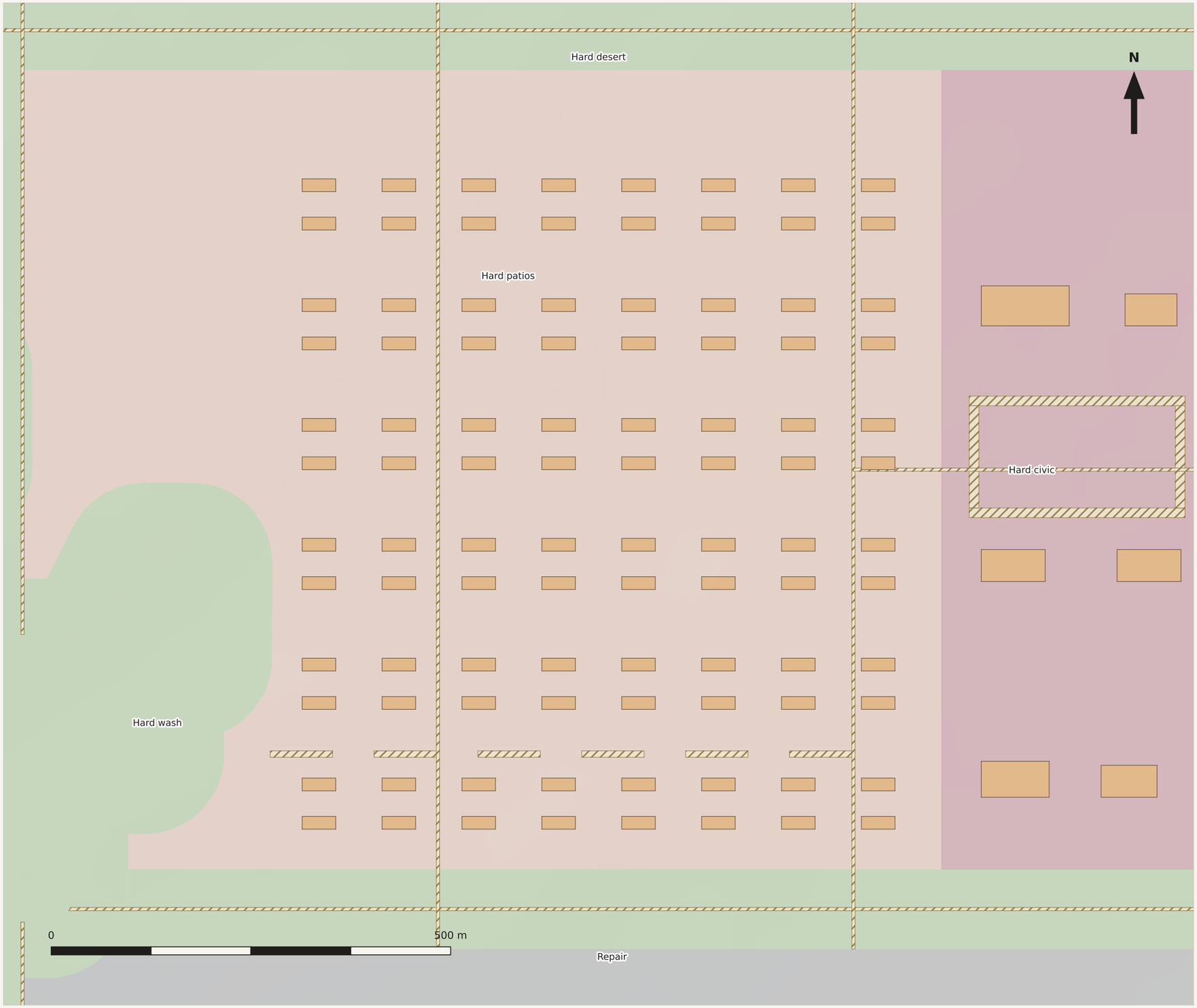
Recomputed

Site coverage 0.40% · FAR 0.01 · GFA 311,343 m²

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Floors from buildings.geojson; metrics recomputed



Legend

- 1 floor
- 2-3 floors
- 4-6 floors
- 7+ floors
- Master design area

Land use in the master area (ha)

reserve 2,838
open space 706
compute 104
residential 103
energy 96
agriculture 82
industry 66
civic 45

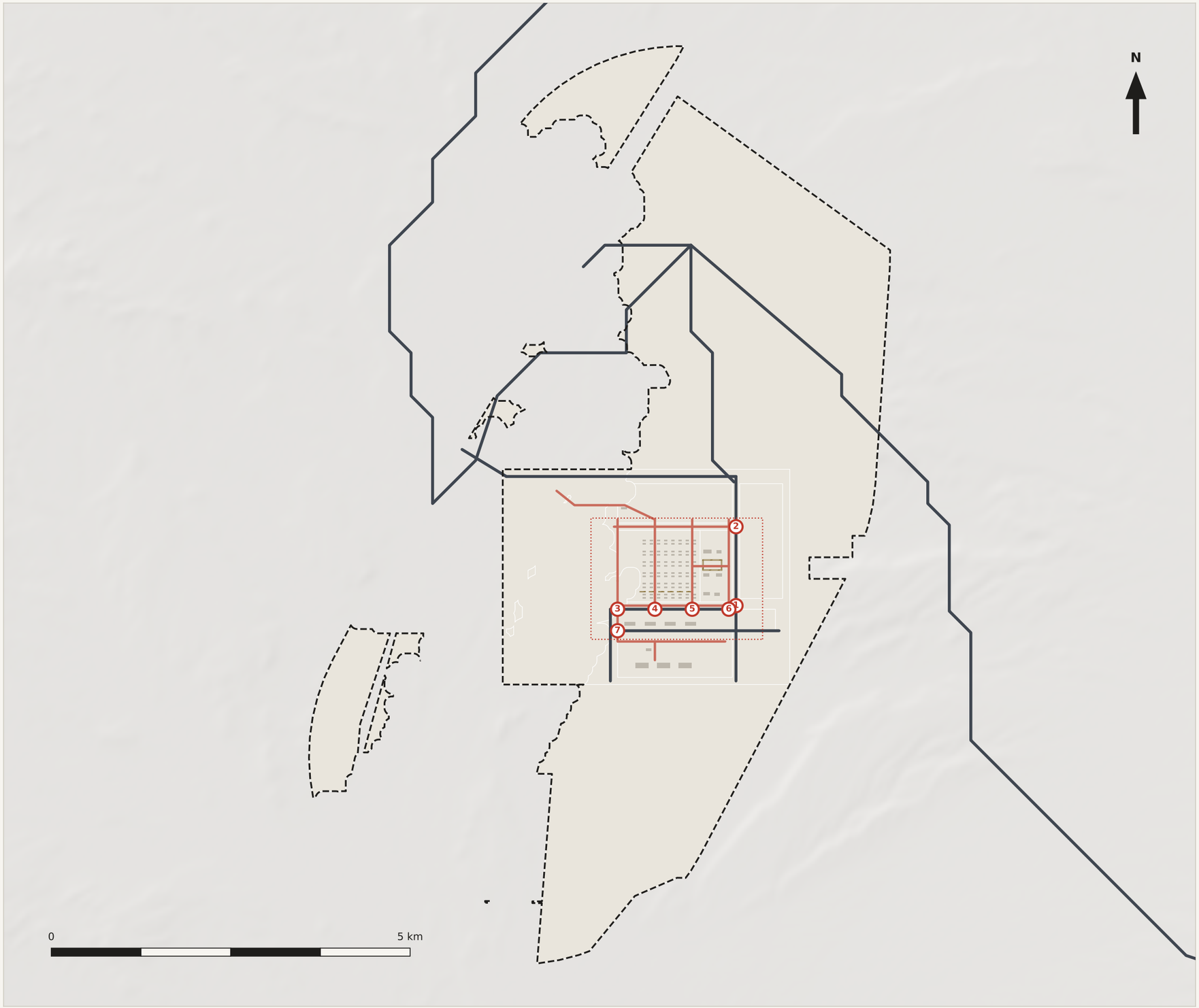
Recomputed from geometry

Coverage 0.40% · FAR 0.01 · open space 18%

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Every human-machine interface numbered



Legend

- People 13 km
- Machines + goods 158 km
- Shared 0 km
- Human-machine interface (7)

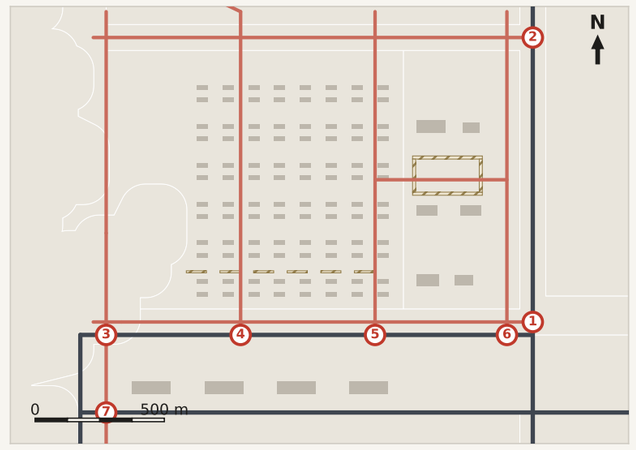
Machine traffic rule

separated

Interfaces

7 at-grade points where people and machines meet

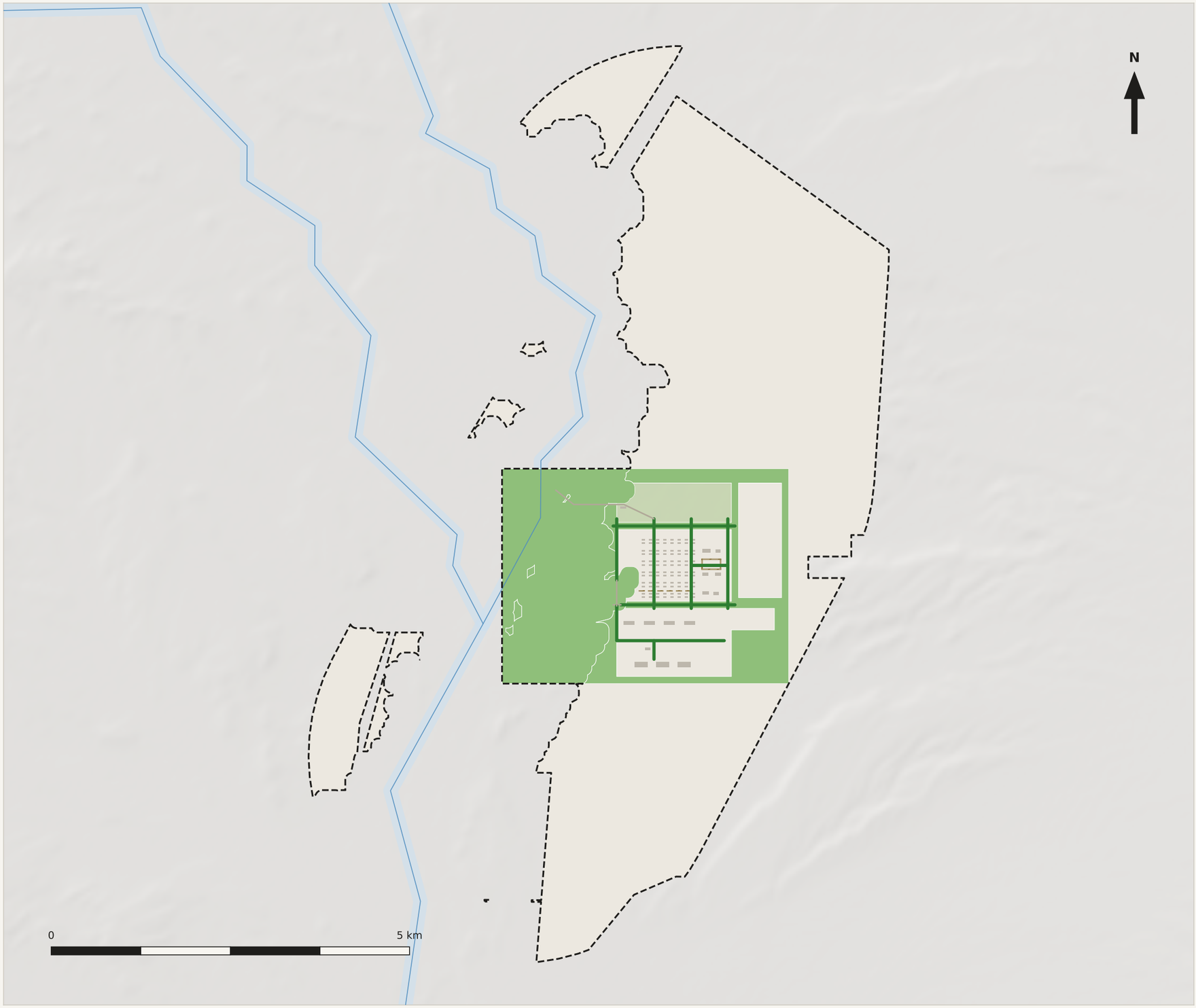
Interfaces, enlarged



Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Channels are DEM-derived flow lines



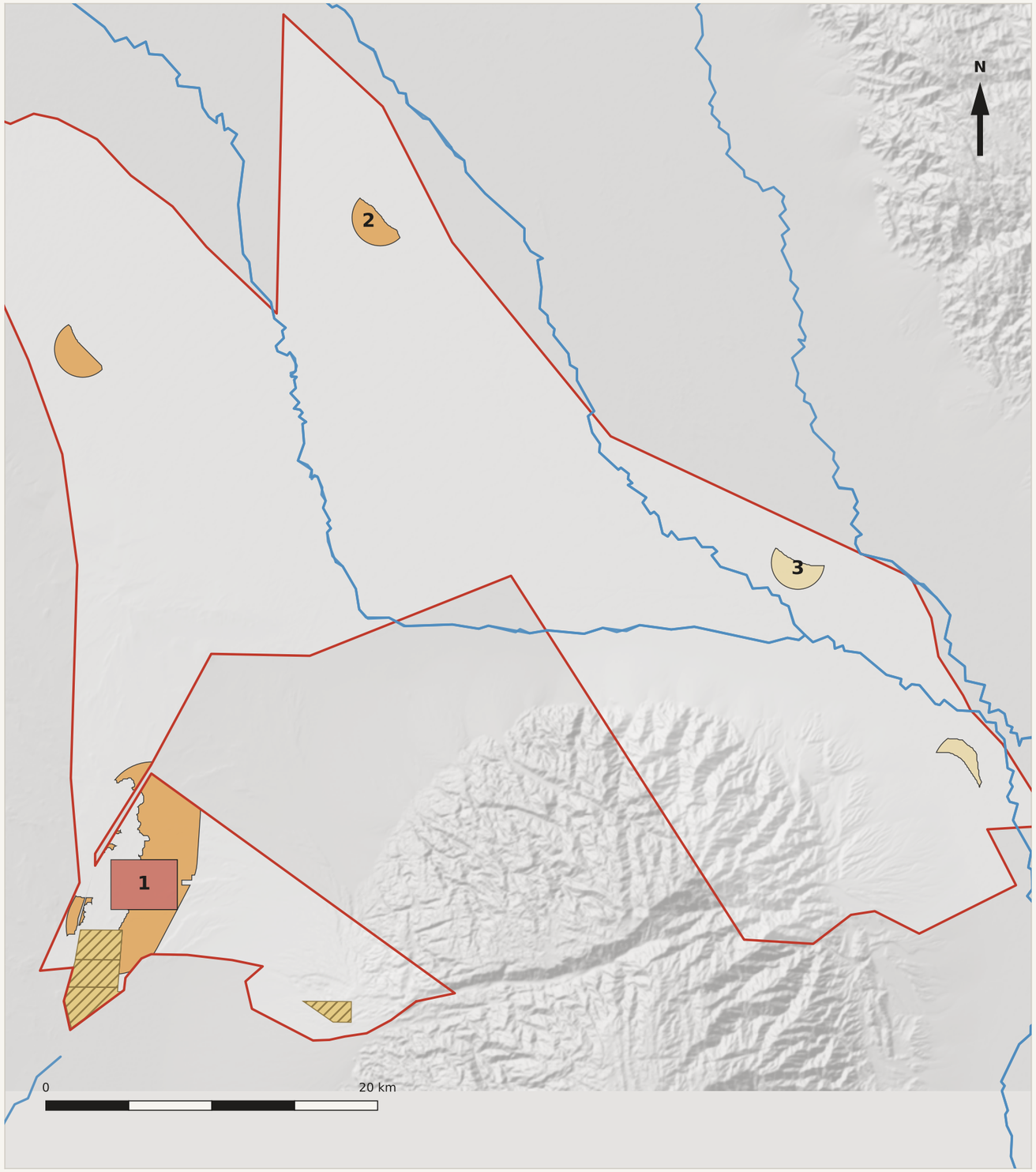
Legend

- Open space 18%
- Agriculture
- Shaded walking 84%
- Unshaded walking
- Seasonal channel
- Flood corridor shown at 250 m / 100 m

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Observable triggers, not dates



Legend

- Phase 1 12 km²
- Phase 2 38 km²
- Phase 3 7.21 km²

Phase 1 - Independent first neighbourhood and regional access works

Phase one: no trigger.

Phase 2 - Conditioned urban and productive expansion

Twelve months audited 20% reliable-water headroom; 80% occupancy; each extension has funded services, processing and safe access

Phase 3 - Eastern gateway and industry

Hydraulic and road engineering accepted; binding trade demand; lawful reliable water and power; own services and emergency access funded

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Every declared figure with its evidence tag

Metric	Declared	Unit	Evidence	Basis / check
master_area_km2	40	km2	computed	recomputed 40
site_coverage_pct	0.40	%	computed	recomputed 0.40
gfa_m2	311,343	m2	computed	recomputed 311,343
far	0.01	ratio	computed	recomputed 0.01
open_space_pct	17	%	computed	recomputed 17
human_machine_interfaces	7	count	computed	recomputed 7
shaded_pedestrian_pct	84	%	computed	recomputed 84
structure_km2	per system	km2	computed	recomputed: per system
hourly_cfe_match_pct	62	%	computed	tools/hourly_dispatch.csv
peak_it_load_mw	12	MW	assumed	A-COOL
annual_water_demand_m3	1,000,000	m3/yr	assumed	A-WATER
water_domestic_m3	140,160	m3/yr	assumed	A-RATES
water_compute_m3	1,577	m3/yr	assumed	A-COOL
water_agriculture_m3	778,263	m3/yr	assumed	A-RATES
water_landscape_m3	20,000	m3/yr	assumed	A-RETURN
water_industry_m3	60,000	m3/yr	assumed	A-RETURN
wue_l_per_kwh	0.02	L/kWh	assumed	A-COOL
pue	1.25	ratio	assumed	A-COOL
water_reuse_pct	20	%	assumed	A-RETURN
waste_heat_reuse_pct	5	%	assumed	A-HEAT
jobs_phase1	1,200	jobs	declared	
residents_phase1	2,400	persons	declared	
dwellings_phase1	960	units	declared	
machine_traffic_rule	separated	enum	declared	

Evidence tags

- computed - recomputed from your geometry
- sourced - cited in sources.json
- assumed - declared in assumptions.json
- indicative - organiser's indicative material

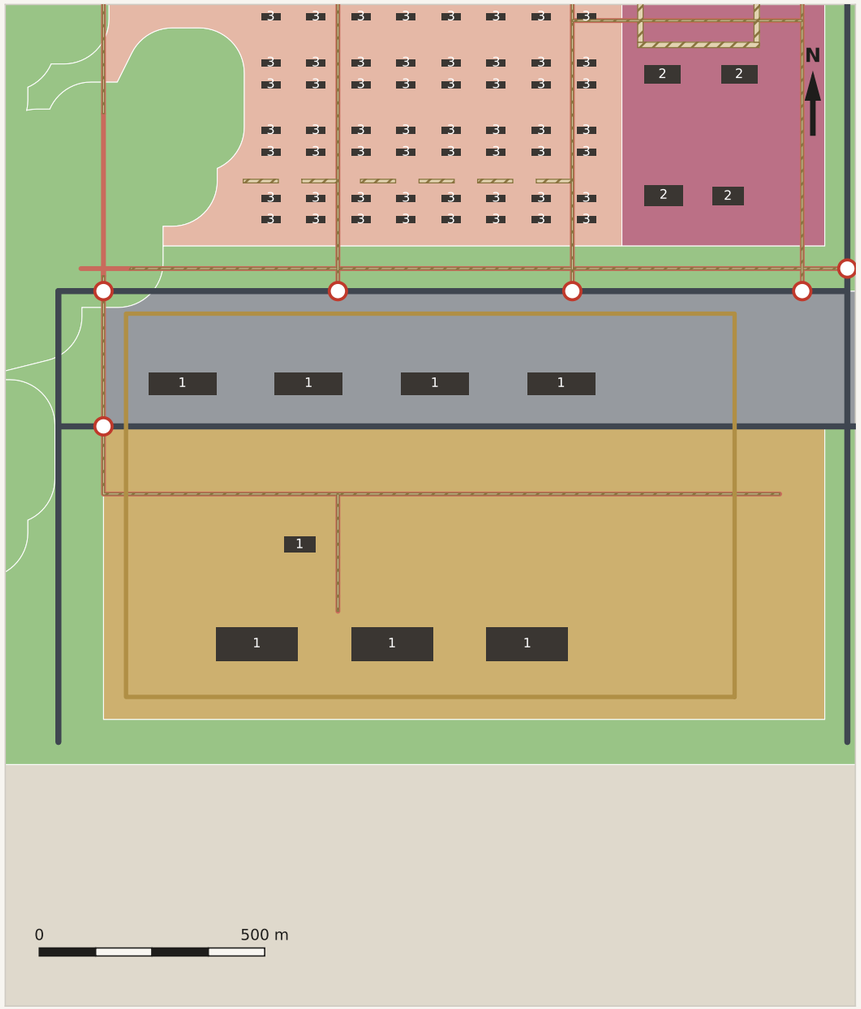
Gate G4

Every metric tagged computed is recomputed from geometry/ by the self-check; a declared value more than 2% away fails.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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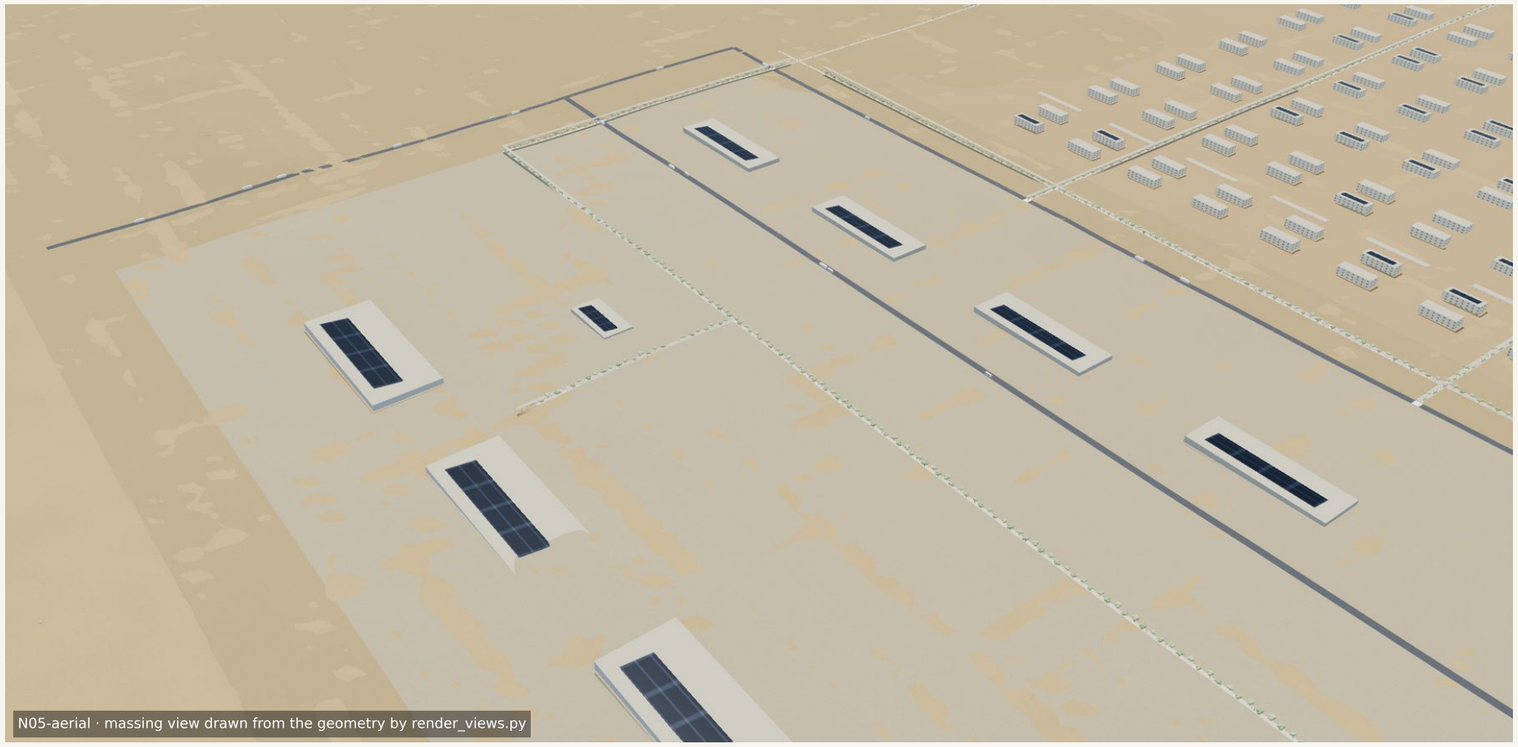
Plan · floors marked on each building · interfaces circled



Heat without thirst

Dry cooling and cooperative herb drying share a metered heat loop

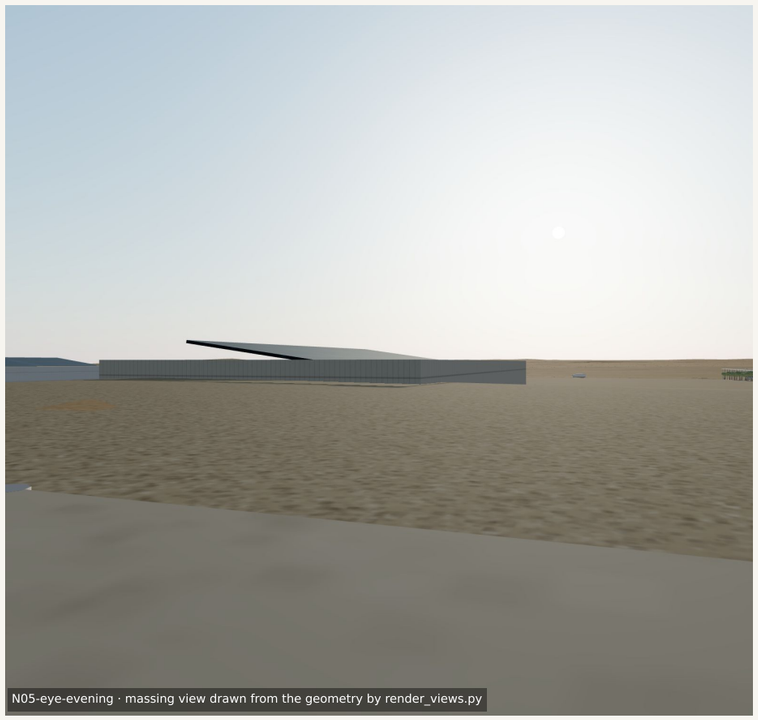
Heat without thirst



N05-aerial · massing view drawn from the geometry by render_views.py



N05-eye-day · rendered with OpenAI built-in image generation; exact model version not exposed from the massing view N05-eye-day · illustrates the geometry, not evidence

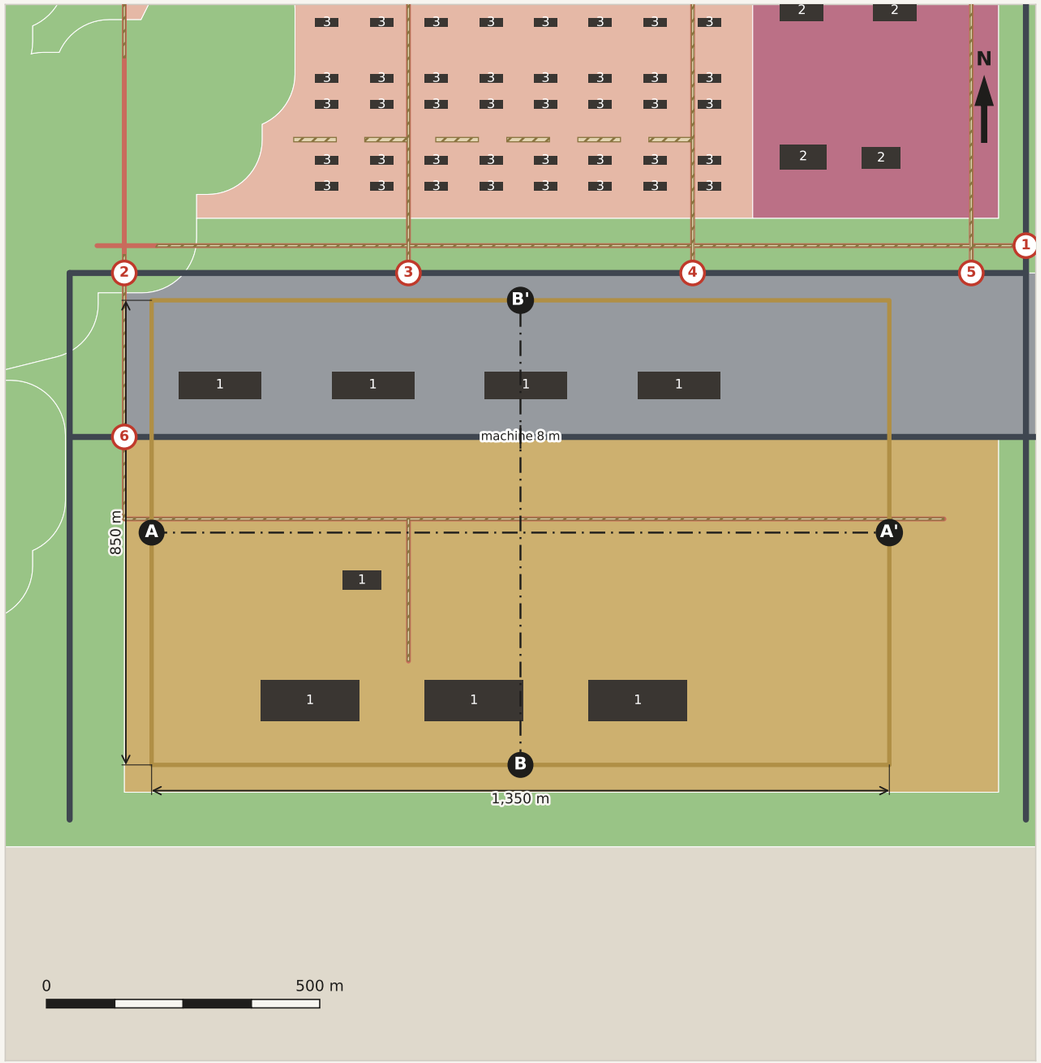


N05-eye-evening · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

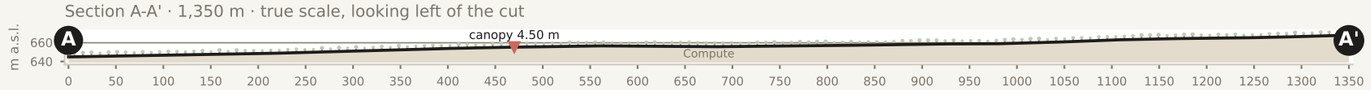
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Plan · overall dimensions, street widths, interfaces numbered, section lines

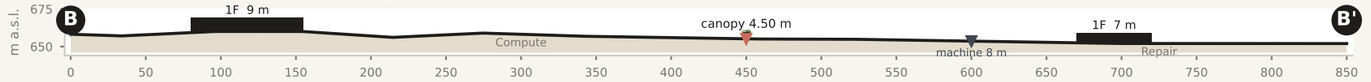


Scale 3 · dimensions and sections from the geometry

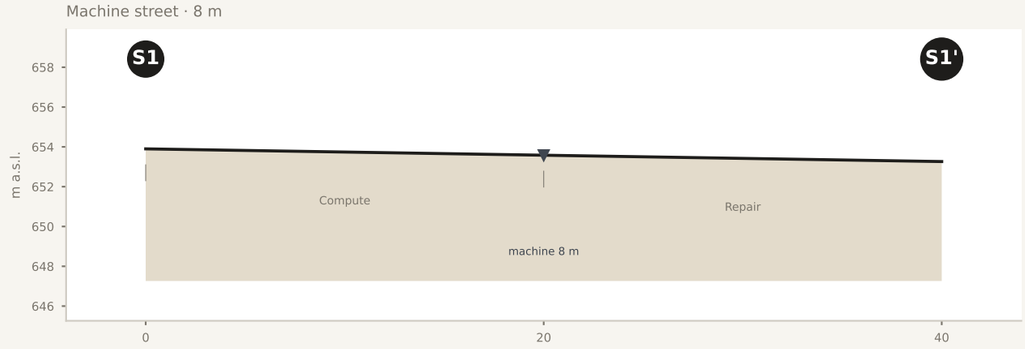
Section A-A' · 1,350 m · true scale, looking left of the cut



Section B-B' · 850 m · true scale, looking left of the cut



Machine street · 8 m



Numbers in the node · recomputed from the geometry

Node area 1,147,500 m² (115 ha)
Buildings 8 · footprint 72,950 m² · floor area 72,950 m²
Coverage 6.40% · FAR 0.06 · open space 0%
Floors: 8 x 1F
Floor area by use: compute 40,500 m², industry 32,500 m²
Canopies (roof on posts, no floor area) 6,440 m²
Networks in the node: machine 1,350 m, people 1,610 m
Shaded share of the walking network 100%
Human-machine interfaces 0
Trees 175 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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A day here

	People	Machines
06:00	Morning market and shaded walk	Freight remains in service yard
14:00	Cooled refuge opens; wash closes on flood warning	Flexible compute sheds load
21:00	Kitchen and sheltered paths remain open	Deliveries transfer at gates

Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



Interfaces in view

1 Gated crossing 3
2 Gated crossing 4
3 Gated crossing 5
4 Gated crossing 7

People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

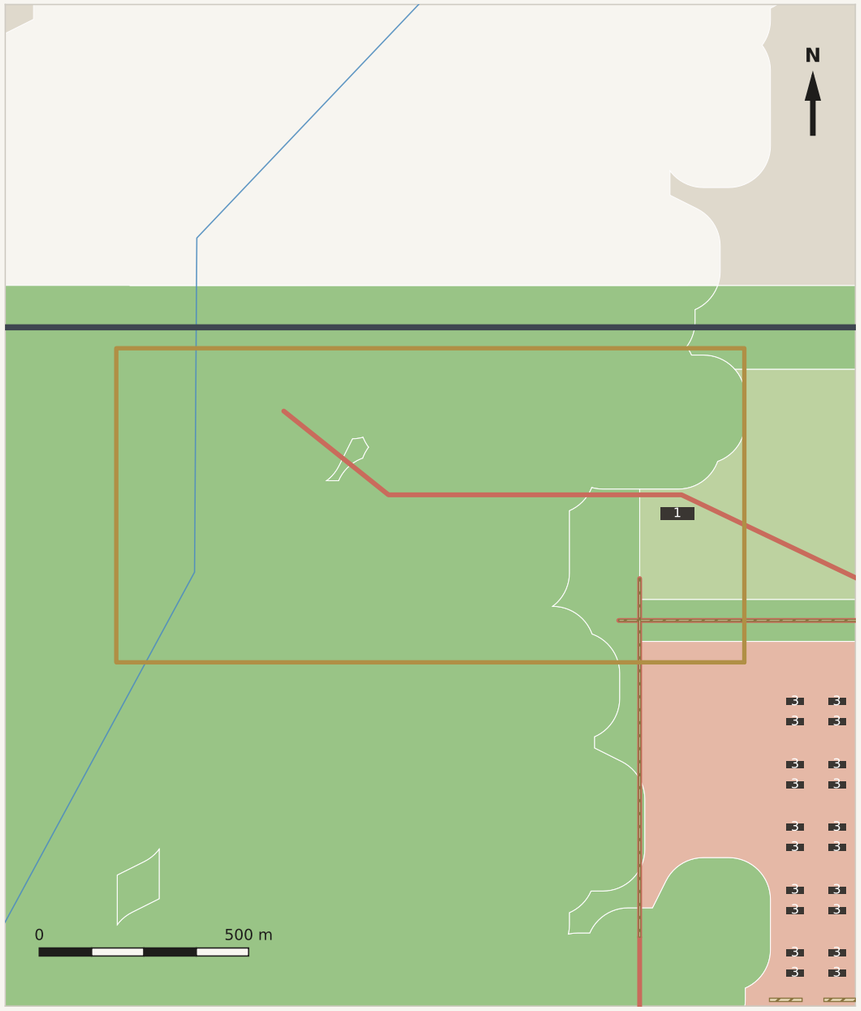




Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Plan · floors marked on each building · interfaces circled



The absent river

Dry channel stays open and paths retreat to high ground

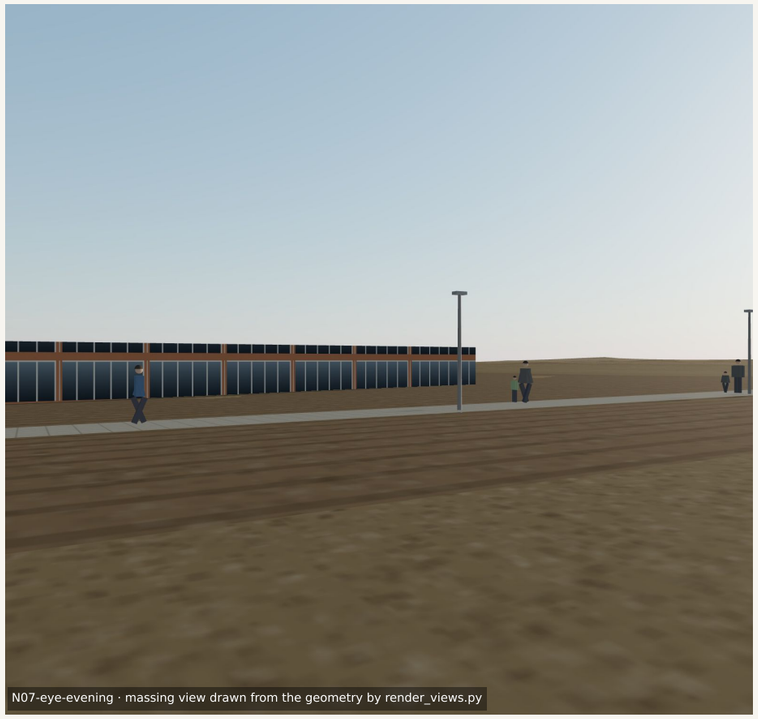
The absent river



N07-aerial · massing view drawn from the geometry by render_views.py



N07-eye-day · rendered with OpenAI built-in image generation; exact model version not exposed from the massing view N07-eye-day · illustrates the geometry, not evidence

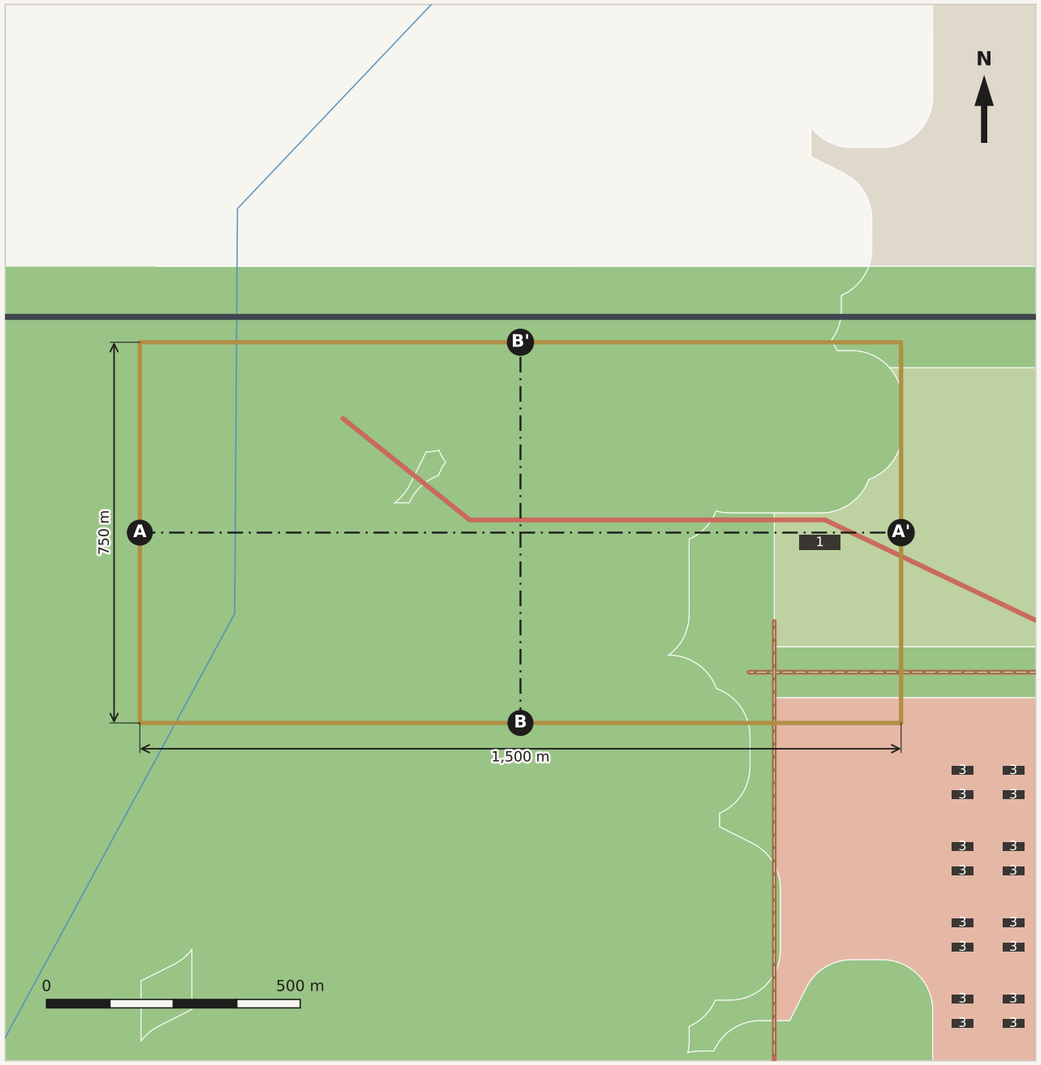


N07-eye-evening · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

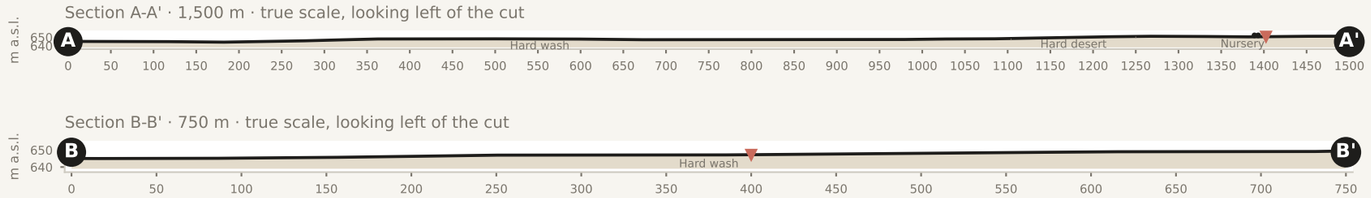
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Plan · overall dimensions, street widths, interfaces numbered, section lines



Scale 3 · dimensions and sections from the geometry

Section A-A' · 1,500 m · true scale, looking left of the cut



Section B-B' · 750 m · true scale, looking left of the cut

Numbers in the node · recomputed from the geometry

Node area 1,125,000 m² (112 ha)
Buildings 1 · footprint 2,400 m² · floor area 2,400 m²
Coverage 0.20% · FAR 0 · open space 92%
Floors: 1 x 1F
Floor area by use: civic 2,400 m²
Canopies (roof on posts, no floor area) 2,000 m²
Networks in the node: people 1,686 m
Shaded share of the walking network 30%
Human-machine interfaces 0
Trees 55 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

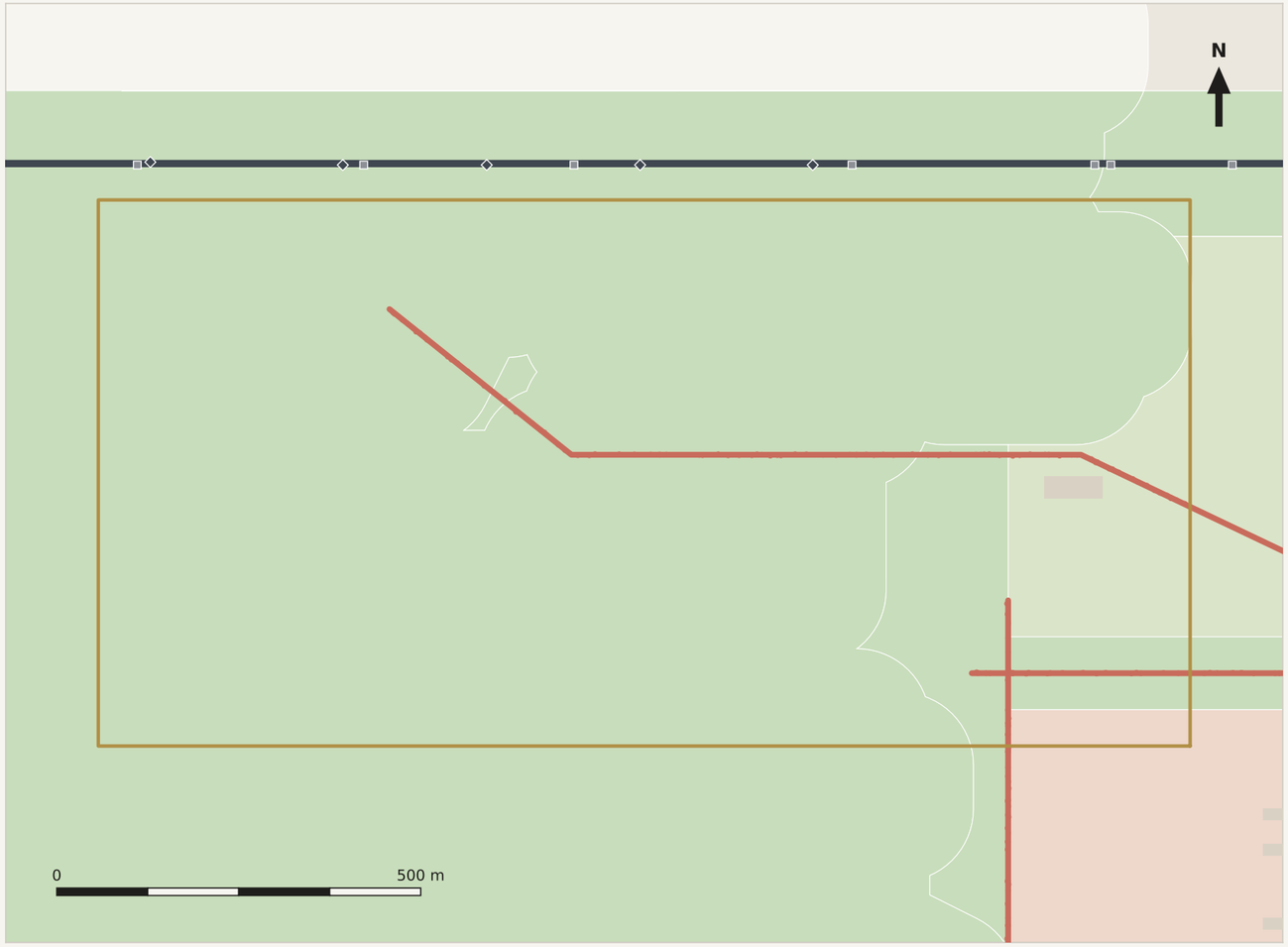
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A day here

	People	Machines
06:00	Morning market and shaded walk	Freight remains in service yard
14:00	Cooled refuge opens; wash closes on flood warning	Flexible compute sheds load
21:00	Kitchen and sheltered paths remain open	Deliveries transfer at gates

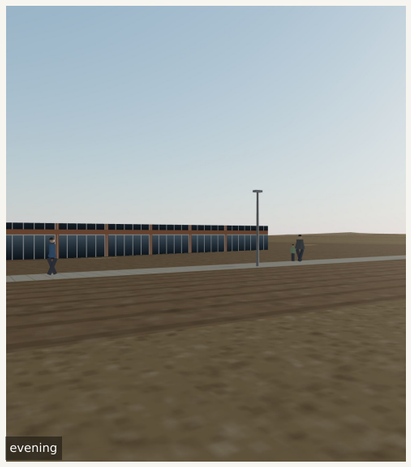
Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface

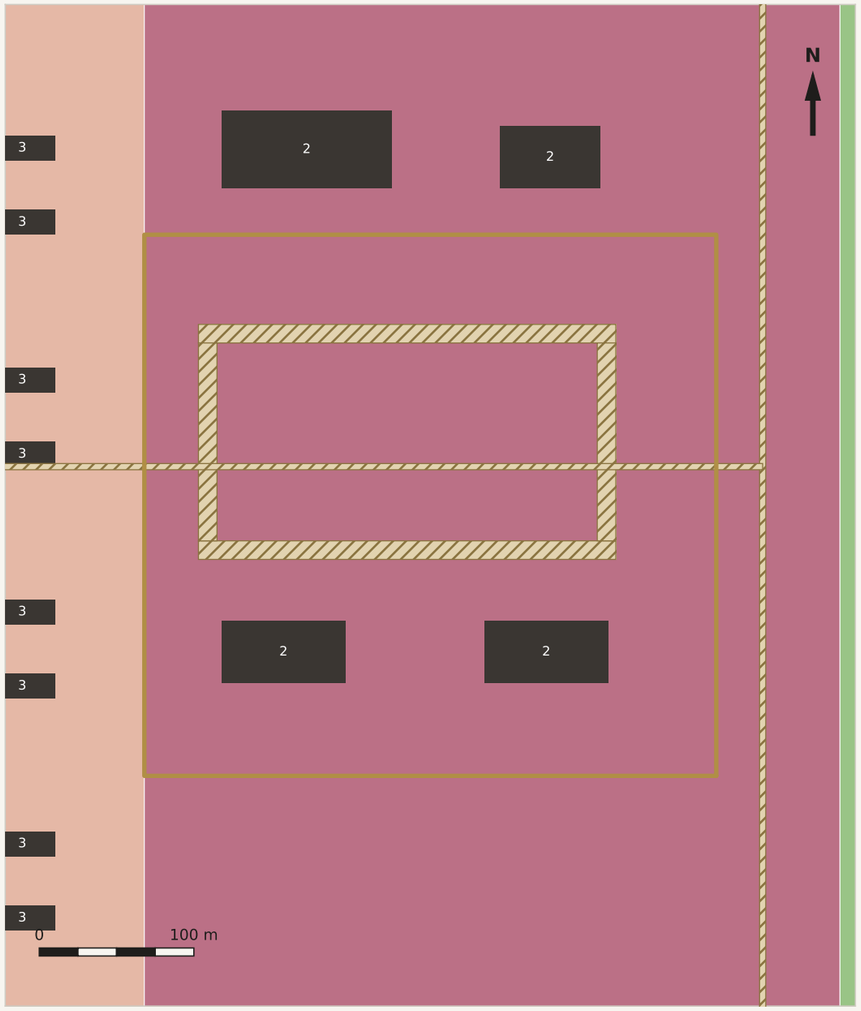




Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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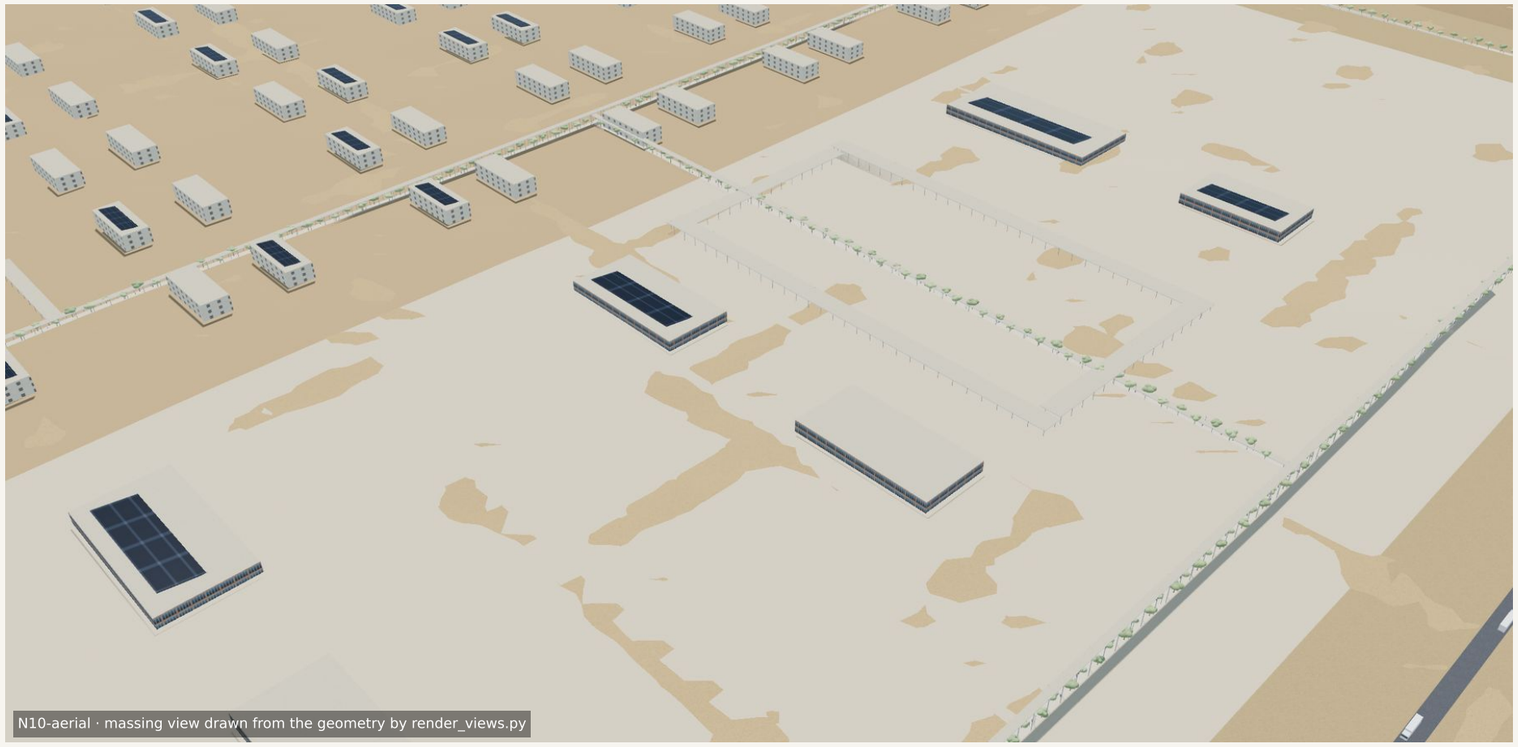
Plan · floors marked on each building · interfaces circled



The right to shade

Public kitchen and library make a commons without ornamental water

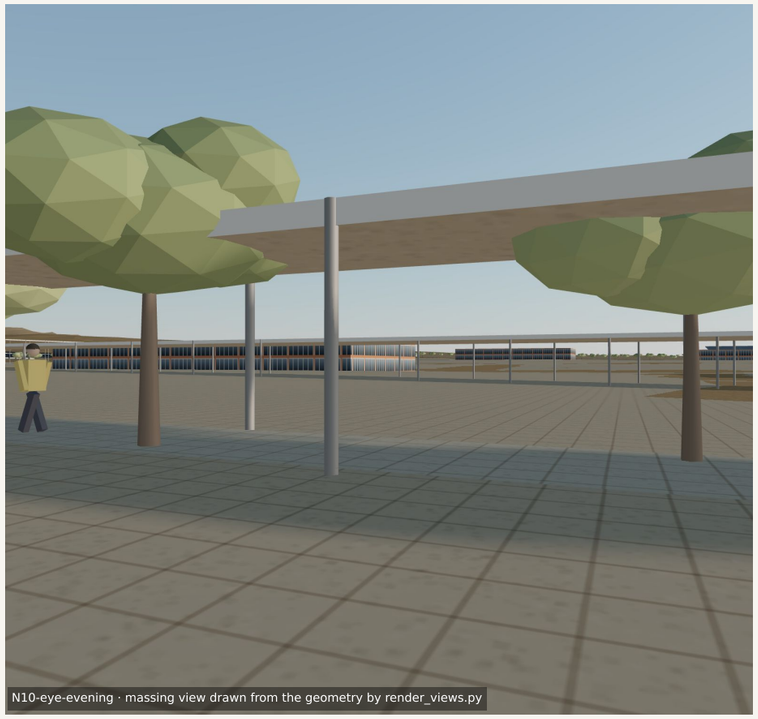
The right to shade



N10-aerial · massing view drawn from the geometry by render_views.py



N10-eye-day · rendered with OpenAI built-in image generation; exact model version not exposed from the massing view N10-eye-day · illustrates the geometry, not evidence

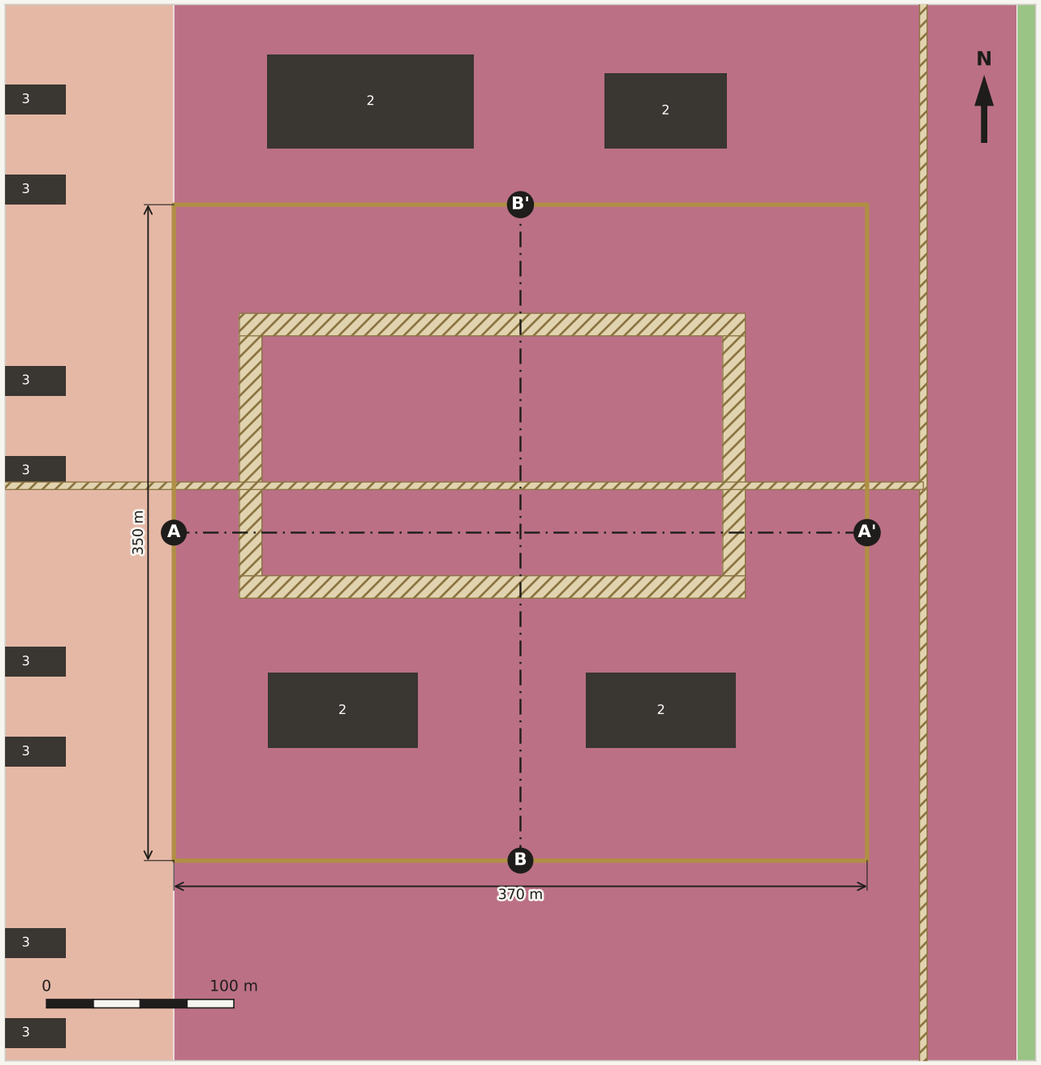


N10-eye-evening · massing view drawn from the geometry by render_views.py

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

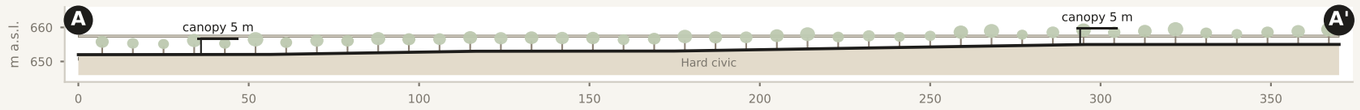
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Plan · overall dimensions, street widths, interfaces numbered, section lines

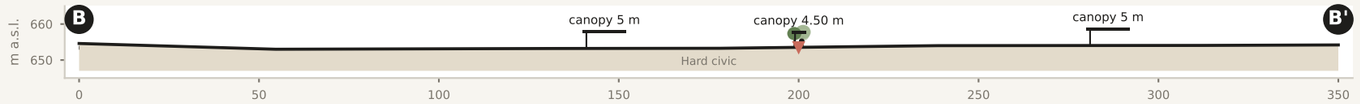


Scale 3 · dimensions and sections from the geometry

Section A-A' · 370 m · true scale, looking left of the cut



Section B-B' · 350 m · true scale, looking left of the cut



Numbers in the node · recomputed from the geometry

Node area 129,500 m² (13 ha)
Buildings 2 · footprint 6,400 m² · floor area 12,800 m²
Coverage 4.90% · FAR 0.10 · open space 0%
Floors: 2 x 2F
Floor area by use: civic 12,800 m²
Canopies (roof on posts, no floor area) 11,030 m²
Networks in the node: people 370 m
Shaded share of the walking network 100%
Human-machine interfaces 0
Trees 41 (as placed in the 3D scene)

Cut buildings in black (floors x 3.5 m, or height_m); beyond the cut, buildings and trees within 45 m in grey; trees and people as placed in the 3D scene; terrain from the DEM.

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

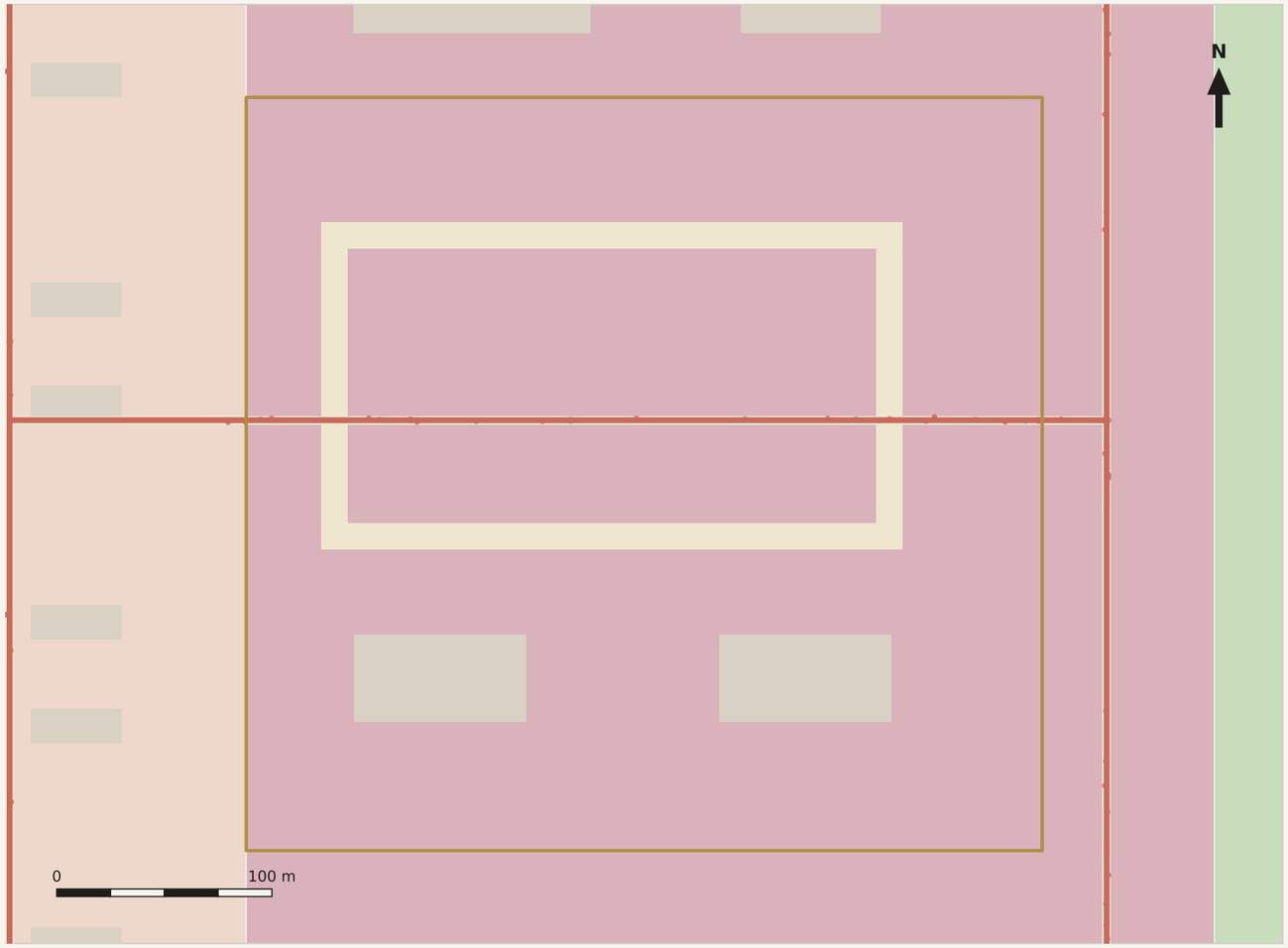
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A day here

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06:00	Morning market and shaded walk	Freight remains in service yard
14:00	Cooled refuge opens; wash closes on flood warning	Flexible compute sheds load
21:00	Kitchen and sheltered paths remain open	Deliveries transfer at gates

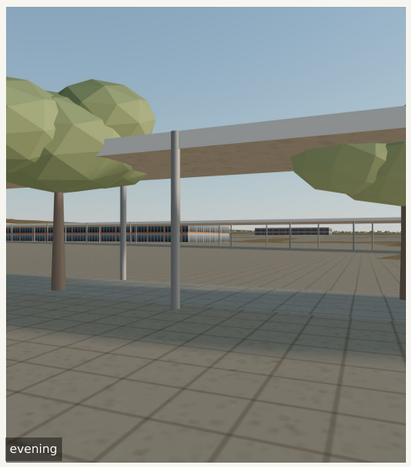
Who and what moves here: networks, interfaces, and the people, robots and shuttles of the 3D scene

People and machines through one day



People
Machines + goods
Shared at walking pace
People (scene)
Delivery robots
Driverless shuttles
Human-machine interface





Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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The same places, used through a day



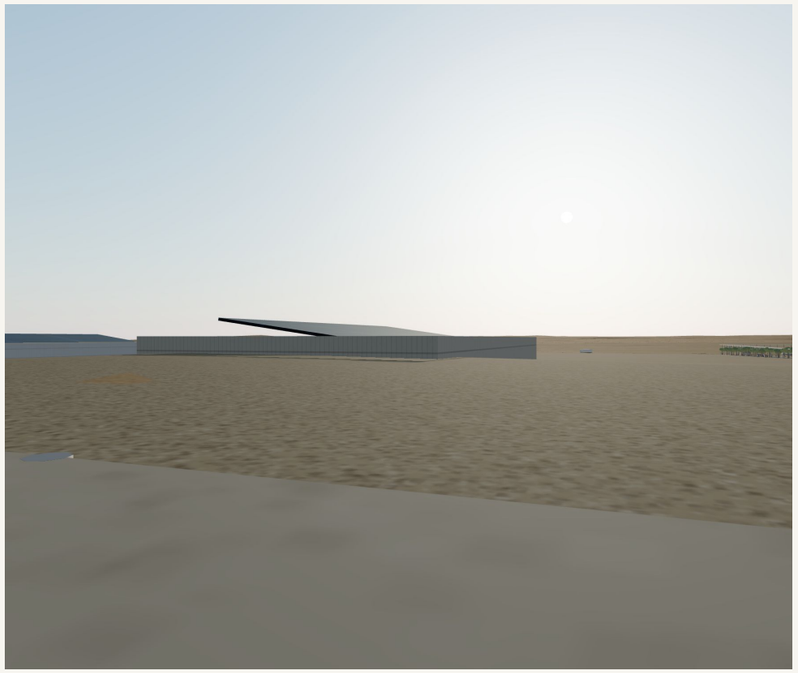
N05 A compute campus and its waste heat · day



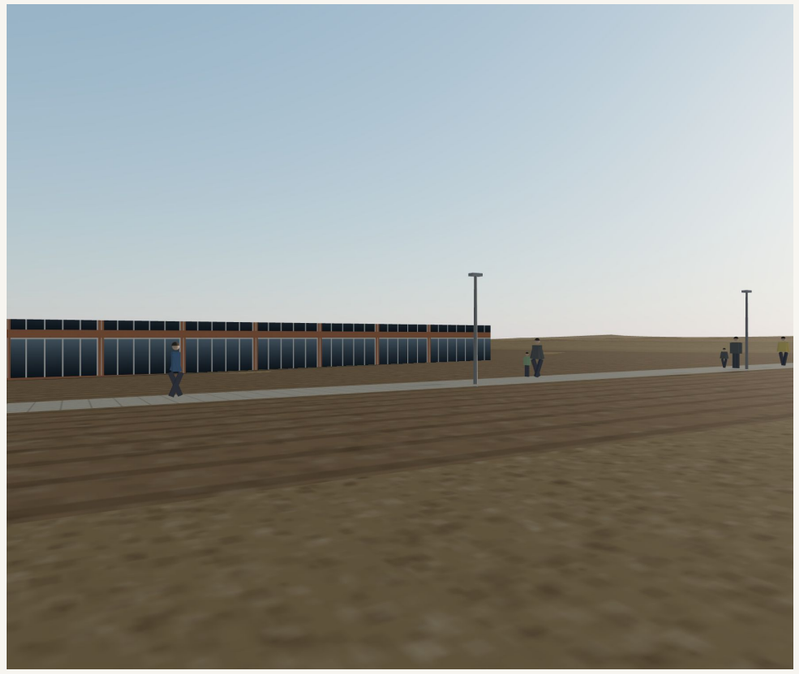
N07 A seasonal river park · day



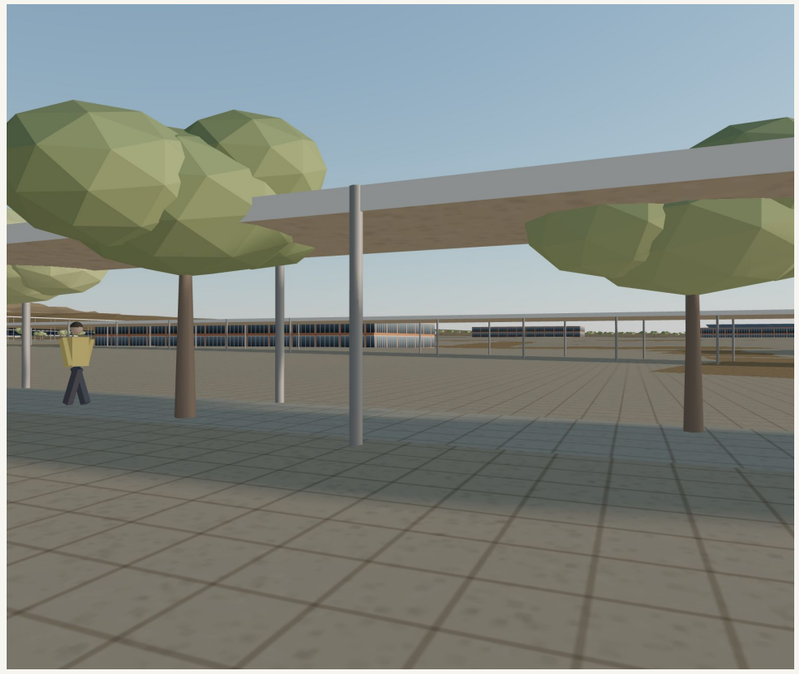
N10 Public space in heat and Zonda wind · day



N05 A compute campus and its waste heat · evening



N07 A seasonal river park · evening



N10 Public space in heat and Zonda wind · evening

Sources: parcels and pilot area - XCITY (owner KML); terrain and channels - produced using Copernicus WorldDEM-30 (c) DLR e.V. 2010-2014 and (c) Airbus Defence and Space GmbH 2014-2018 provided under COPERNICUS by the European Union and ESA; all rights reserved; roads (c) OpenStreetMap contributors. Everything else: this submission's own geometry.

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Assumptions

A-WATER Water envelope: 1,000,000 m3/year gross use: 840,000 new freshwater plus 200,000 internally reclaimed water; not an awarded abstraction right. 40,000 m3/year upstream distribution loss; 800,000 fresh reaches uses. - No supplied water right or supply design. Reuse is an internal transfer. Construction requires lawful reliable fresh supply.

A-PILOT Existing pilot accounting: 600,000 m3/year of agriculture envelope provisionally reserved for existing pilot; owner metering must replace this before operation. No fixed arrays, plants or grazing arrangements changed. - Existing use is unmetered here. Higher verified demand reduces new discretionary uses first; existing operations remain unchanged.

A-RATES Domestic and new agriculture: 2,400 residents at 160 L/person/day = 140,160 m3/year; new irrigation 170,000 m3/year; panel washing 8,263.2 m3/year. Total agriculture including pilot 778,263.2 m3/year. - Assumed rates, not local measurements. New irrigation is capped at 30 ha plus nursery; reserved farmland is not all irrigated.

A-COOL Cooling basis: Average IT 9 MW (12 MW for 12 hours, 6 MW for 12 hours); annual IT 78,840 MWh. PUE 1.25; site WUE 0.02 L/kWh; annual compute water 1,576.8 m3. - Closed-loop chip cooling with dry heat rejection and air-cooled mechanical trim; WUE includes makeup and facility uses, no evaporative tower. Performance targets require supplier confirmation at 44 C.

A-ENERGY Hourly energy scenario: 40 MW PV; synthetic annual capacity factor 25%; 480 MWh battery, 75 MW converter, 95% efficiency each direction; 216 MWh delivered reserve held for 72 hours at 3 MW critical load. Grid import capacity assumed 30 MW; no export. - Illustrative 8760-hour sinusoidal profile, 22% seasonal variation and three dust days each 60 days. No site weather series, tariff or connection offer supplied. tools/energy.py is reproducible, not validated forecasting.

A-RETURN Reuse mass balance: 200,000 m3/year treated reuse: 100,000 domestic return + 50,000 industrial return + 50,000 captured agricultural drainage; dispatched 180,000 agriculture and 20,000 landscape. Of annual 840,000 fresh input, 40,000 is distribution loss and 800,000 terminal consumption/losses. - Assumed recoverable returns require treatment, disinfection and salinity checks. No rain or condensate yield credited. Concentrates require contained handling and licensed removal.

A-SHORT 30 percent gross-use reduction: From 1,000,000 to 700,000 m3/year: landscape 20,000 cut; new irrigation and washing 178,263.2 cut; industry 40,000 cut; domestic/community nonessential use 61,736.8 cut. Existing pilot 600,000 and compute makeup 1,576.8 protected. - Gross-use shortage specified by brief, distinct from fresh-water shortage. Domestic/community residual is 78,423.2 m3/year (89.52 L/person/day including communal uses), an emergency assumption requiring verified essential-use sufficiency; if insufficient, secure emergency import or defer occupancy rather than deprive essential use.

A-FLOOD Screening buffers: Existing permanent buildings retain conservative exclusion of mapped flood band plus 30 m and channels plus 150 m. Whole-site flood-sensitive screening is shown separately from mountain/flow corridors; it is not a confirmed legal no-build boundary. - Screening offsets, not hydraulic standards. Minor washes and external inflows are missing; ground survey and 2D modelling required.

A-SEISMIC Structural concept: One to three storeys; regular reinforced concrete frames and ductile walls, braced light roofs; conceptual separation gaps 100 mm minimum subject to drift calculation; design reference INPRES-CIRSOC 103 Part I 2018 with applicable addenda to be confirmed by local engineer. - No soil class, spectra or geotechnical report supplied. Geometry is urban design, not structural certification.

A-ACCESS Access allowance: Retire the arbitrary 80 km external road allowance. Regional RN40 approach shown as unverified directional context; internal candidate corridors are measured in the V3 GeoJSON. Actual external route, easements and cost remain phase-one prerequisites. - Actual route, easements and crossings unresolved. The nearest-site distance is not the settlement access length. Phase one requires funded access.

A-SHADE Heat and light: Shade share counts actual proposed roof coverage; thermal comfort requires testing. 2200 K shielded night lighting; no facade uplighting; 10 lux route design target subject to engineering. - No outdoor comfort simulation or illuminance survey available. Enclosed cooled refuges are required at 44 C; shade is not a guarantee of safe conditions.

A-HOUSING Population and employment: 960 dwellings; 2.5 persons per dwelling = 2,400 residents; 1,200 local jobs. 1,080 resident workers, 120 inbound specialists; 120 resident remote workers, 600 children/students, 300 retired adults, 300 unpaid carers or job-seekers. - Programme targets, not labour-market estimates. Resident categories sum to 2,400; remote-worker income, pensions and household transfers support nonlocal-job households.

A-HEAT Heat receivers: Drying cooperative accepts annual 3,942 MWh usable heat, equal to 5% of IT electricity, via a metered insulated loop and temperature-lift heat pump if required. - Declared contracted-use target; herb drying and process wash-water preheat are named receivers, not assumed continuous demand. Reject all unused heat to air; no winter greenhouse demand invented.

A-GROWTH Growth ceiling: Four conditional town reservations A/B/C/D plus the fixed first living quarter; only the detailed first neighbourhood has 960 proposed dwellings / 2400 residents. No 7200-person whole-site ceiling or final regional population is asserted. - A deliberately modest ceiling, not a population forecast. Reservations confer no development entitlement.

A-COST Funding order: Infrastructure escrow and 12 months public-service operating reserve precede dwelling occupation; compute concession pays annual fixed community service charge independent of runtime. - No prices or financing terms supplied; these are proposed enforceable contract conditions, not existing agreements.

A-V3 Whole-site planning screen: 300m block-mean DEM; mountain/steep screen slope >=5% or elevation >=850m; channel research corridor 150m each side; four towns and two compute reservations clipped by conservative flood and terrain screens. - Regional design choices, not statutory restrictions; the source flood band is a sensitivity screen. Eastern gateway strict-screen candidate about 2.64 km2 remains below the brief's about 7km2 intent. Regional capacity, fixed living-quarter demand, water sources, road engineering and resources remain unverified.

Sources

SRC-KML Parcel boundaries for parcels 9 and 13 - XCITY (owner-supplied KML)
https://github.com/XcityUS/xcity-ai-competition

SRC-PILOT Fixed agrivoltaic fields and first living quarter - XCITY
https://github.com/XcityUS/xcity-ai-competition

SRC-DEM Copernicus DEM GLO-30 (WorldDEM-30) - European Union / ESA, Copernicus programme
https://github.com/XcityUS/xcity-ai-competition

SRC-OSM OpenStreetMap road network - OpenStreetMap contributors
https://github.com/XcityUS/xcity-ai-competition

SRC-XCITY-INDICATIVE Indicative positions of the five urban systems - XCITY
https://github.com/XcityUS/xcity-ai-competition

S-DOE Cooling Water Efficiency Opportunities for Federal Data Centers - US Department of Energy https://www.energy.gov/cmei/femp/cooling-water-efficiency-opportunities-federal-data-centers

S-INPRES Reglamentos INPRES-CIRSOC - INPRES
https://www.inpres.gob.ar/Ing_Sismorres/Reglamentos.html

S-CFE 24/7 Carbon-Free Coalition technical guidance - Climate Group
https://www.theclimategroup.org/247-technical-guidance

AI disclosure

Agent Codex, model GPT-6 family (serving version unavailable), runtime Codex desktop / macOS; local Python and offline Three.js. Autonomy: human_directed. Human selected Finite Water City and the three nodes, confirmed jingyuanhuang-design / Jingyuan Huang, requested clearer overall layout, reviewed the V2 diagrams, and then explicitly requested public publication on 2026-10-25. Codex authored geometry, calculations, text and visuals. Public preview publication is authorised; formal competition entry awaits the organiser opening submissions. The new V2 walking spine and arrival point remain explicitly identified proposals. On 2026-10-26 the human requested a whole-site redesign under the updated brief, questioned the excessive flood reserve depiction and directed Codex to revise only this entry while other agents handle repository rules. V3 distinguishes flood screening from permanent restrictions. The human has not yet reviewed the integrated V3 package. Human requested an architectural rendering cover instead of the abstract slogan on 2026-10-26; a

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AGUA FINITA | The Finite Water City　有限水城

first-neighbourhood aerial was produced for the cover.

Images

9 images registered in image_provenance.json. Views in this booklet are, in order of preference: an image made with an image model from a massing view of the geometry (model, prompt and control image registered), the massing view itself (scripts/render_views.py), or a massing drawing by scripts/make_figures.py. All of them explain the geometry; none is evidence.

OpenAI built-in image generation; exact model version not exposed - 6 images, post-processing none. Files in media/: N05-eye-day.png (control N05-eye-day.png), N07-eye-day.png (control N07-eye-day.png), N10-eye-day.png (control N10-eye-day.png), city-aerial.png (control city-aerial.png), city-evening.png (control city-evening.png), cover-render-v5.png (control city-aerial.png). Full prompts in image_provenance.json.

Python matplotlib / Shapely; deterministic code drawing, no image-generation model - 3 images, post-processing Regional 300m terrain screening and labelled vector drawing; diagram layout only. Files in media/: 01_whole_site_plan.png (control -), 02_corridors_and_phases.png (control -), 03_source_geometry.png (control -). Full prompts in image_provenance.json.

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